

Incentives Design, Incentive Compatibility and the Measurement of Inflation Expectations^{*}

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Abstract

Inflation expectations play a central role in macroeconomic models and the conduct of monetary policy. This paper examines how incentive design affects the measurement of inflation expectations. We compare more complex, fully incentive-compatible scoring rules with simpler, non-incentive-compatible incentive schemes in an experimental survey setting. Incentives matter for reported expectations: performance-based payments reduce upward bias, cross-sectional disagreement, and the frequency of extreme responses. Moreover, simpler incentive schemes outperform more complex mechanisms in external and internal validation tests. Expectations elicited under simpler designs are more strongly associated with spending behavior and display greater consistency between point forecasts and density

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forecasts. Our findings suggest that while complex mechanisms ensure truthful reporting in theory, simpler incentive schemes may produce more reliable and behaviorally meaningful measures of inflation expectations in practice. Our findings have important implications for the estimation and interpretation of macroeconomic models—particularly New Keynesian models—that rely on survey-based measures of expectations.

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1 Introduction

Household inflation expectations are central to modern macroeconomics and monetary policy, yet measuring them reliably remains challenging. In most large-scale surveys, including the New York Fed’s Survey of Consumer Expectations (SCE), respondents receive a flat payment unrelated to forecast accuracy. This flat-fee structure severs the link between effort and reward and gives no incentive to reason at the margin when reporting beliefs. Both mechanisms are fundamental inputs for information processing and economic behavior. As a result, measured disagreement may reflect reporting frictions and low effort as much as genuine heterogeneity in expectations.

This paper builds on [Drobot et al. \(2024\)](#), who show that introducing marginal incentives in an SCE-style environment materially alters reported inflation expectations and improves multiple measures of data quality. We ask a more granular question: if flat fees are to be replaced, *which* incentive design performs best? This question is especially salient given the growing emphasis in economics on improving survey measurement and reducing reporting error in large-scale data collection. Despite this focus, there is limited systematic evidence on incentive design in survey environments, particularly regarding how incentives should be structured and whether more theoretically sophisticated mechanisms outperform simpler, behaviorally transparent alternatives.

Importantly, the three marginal-incentive schemes we test in this paper differ systematically in their incentive compatibility and cognitive complexity, allowing us to separately identify the effects of presentation complexity and the complexity associated with imposing strict properness while holding the underlying economic environment constant.

In total, the experiment includes four groups. In *Control*, respondents receive a fixed bonus unrelated to forecast performance, mirroring the inflation-expectations elicitation mechanism used in the SCE. In *Both*, compensation depends directly on forecast accuracy: point-forecast payments decline by one half for each one–percentage-point increase in forecast error, and

density-forecast payments depend on the probability mass assigned to the bin containing realized inflation, with one task randomly selected for payment. This scheme introduces clear marginal incentives but is not strictly incentive compatible. In *Both+Formula*, monetary payoffs are identical to *Both*, but the scoring formulas are explicitly presented, holding incentives fixed while increasing formalization and instructional complexity. Finally, in *Median*, we implement binarized strictly proper scoring rules (an exponential rule for point forecasts and a generalized quadratic rule for density forecasts) that are incentive compatible under standard assumptions and, under absolute-loss logic, elicit the median of the subjective distribution, but impose a more complex mapping from reports to payoffs.

Ex ante, the ranking of these designs is ambiguous. Transparent but non-proper rules may be behaviorally effective if simplicity enhances comprehension and effective use (Danz et al. 2022). Strictly proper scoring rules may guarantee truthful revelation in theory but prove harder to implement under cognitive constraints (Charness et al. 2021, Abeler et al. 2025).

The comparison between *Both* and *Both+Formula* provides a clean test of presentation complexity: monetary stakes and payoff mappings are identical, and only the explicit formalization of the scoring rule changes. Any difference therefore reflects the behavioral consequences of formalization itself. By contrast, the comparison between *Both* and *Median* isolates a different margin. Relative to *Both*, *Median* introduces strict properness and a well-defined elicitation target, while simultaneously increasing cognitive demands. This design allows us to assess whether the theoretical gains from incentive compatibility and target implementation outweigh the behavioral costs of added complexity.

Across all major outcomes, replacing flat-fee payments with marginal rewards for forecast accuracy improves data quality relative to *Control*. Incentivized respondents report fewer extreme forecasts, lower average inflation expectations (reducing upward bias), forecast closer to professional benchmarks, and reduce cross-sectional dispersion and demographic expectation gaps. These gains operate primarily through increased cognitive effort rather than greater information acquisition: incentivized respondents spend more time on the forecasting

task, while reported search behavior changes only modestly and does not affect the main results. Because many macroeconomic models — particularly New Keynesian models — rely on survey-based measures of expectations, our findings have important implications for their estimation and interpretation.

A validation exercise using data from the SCE shows that incentivized respondents exhibit response patterns similar to experienced, repeat SCE participants, whereas unincentivized respondents resemble first-time participants. We interpret this evidence as indicating that incentives increase attention, reduce cognitive errors and serve as a substitute for experience gained through repeated survey participation.

These findings reinforce and extend [Drobot et al. \(2024\)](#), showing that marginal incentives robustly improve belief elicitation across a broader and more complex set of designs.

Our central comparative result concerns the relative performance of incentive structures. Across a wide range of economically meaningful criteria — including extreme-response reduction, bias correction, external benchmark alignment, spillover effects to non-incentivized beliefs, and cross-question coherence — the simplest design, *Both*, performs at least as well as and often better than its more complex counterparts. The comparison with *Both+Formula* reveals little incremental benefit from formalization, suggesting that added instructional complexity does not improve elicitation quality. The *Median* treatment delivers its key theoretical prediction: under skewed belief distributions, point forecasts align more closely with density medians than means. However, this theoretical success does not translate into systematic dominance on broader applied measures of data quality.

A main takeaway from these results is that for survey practice, behavioral usability appears at least as important as formal sophistication. If the researcher’s objective is to improve belief-elicitation quality in large-scale household surveys, a transparent, low-complexity marginal-incentive scheme like *Both* is seemingly the most robust design choice. These findings highlight a mechanism-design trade-off between incentive compatibility and implementability under cognitive constraints. In this setting, strict properness is useful for specific identifi-

cation targets (e.g., median elicitation), but it is not sufficient for best overall performance when added complexity reduces effective use of incentives.

1.1 Related Literature

Measuring inflation expectations in macroeconomics A large literature examines how inflation expectations are measured in large-scale surveys. [Armantier et al. \(2013\)](#) provide a comprehensive overview of survey-based measures of inflation expectations, including point and density forecasts. They highlight substantial heterogeneity in household expectations and document the sensitivity of responses to survey mode, framing, and question wording. The Federal Reserve Bank of New York’s Survey of Consumer Expectations ([Armantier et al. 2017](#)) further illustrates the role of careful question design and repeated measurement in eliciting consistent expectations data. Recent syntheses by [D’Acunto et al. \(2023\)](#) similarly show that household forecasts exhibit large dispersion, volatility, and systematic demographic patterns.

Several studies document how survey format and sample selection affect reported beliefs. [De Bruin et al. \(2012\)](#) show that small changes in wording can alter how respondents interpret the concept of “inflation,” often shifting responses in predictable ways. [Goldfayn-Frank et al. \(2025\)](#) show that probabilistic forecast of one year ahead inflation highly depend on the question format. [Jonung \(1981\)](#) and [Pfajfar and Santoro \(2008\)](#) demonstrate persistent cross-gender and cross-demographic differences in inflation perceptions and expectations. Together, these contributions underline that survey responses reflect both information and survey design features, suggesting that the specific incentive schemes employed elicit inflation expectations might also shape reports.

Incentives in macroeconomic belief surveys A growing literature examines the role of monetary incentives in eliciting macroeconomic expectations. [Engelberg et al. \(2009\)](#) use proper scoring rules to incentivize professional forecasters and show significant differences between point predictions and subjective probability distributions. Recent experimental work

by [Drobot et al. \(2024\)](#) directly investigates incentivized versus non-incentivized inflation forecasts, finding substantial differences in dispersion and reported uncertainty.

Theoretical incentive compatibility in belief elicitation The marginally incentivized methods for eliciting inflation expectations build on the broader literature on belief elicitation. Proper scoring rules provide incentive compatible methods for eliciting subjective probabilities from risk-neutral agents ([Gneiting and Raftery 2007](#)). However, their practical implementation often requires adaptations to account for risk aversion or for discrete survey interfaces. One such adaptation is the binarized scoring rule ([Hossain and Okui 2013](#)), which converts any proper scoring rule into a binary lottery to ensure truthfulness under a wide range of risk preferences. Other contributions include [Eyting and Schmidt \(2021\)](#) multiple-point prediction mechanism and the choice-based method proposed by [Goldfayn-Frank et al. \(2024\)](#), both of which seek to simplify the reporting space while maintaining theoretical incentive compatibility.

Overall, theoretical incentive compatibility requires that truthful reporting maximize expected monetary payoff under standard assumptions about risk preferences and understanding of the payoff mapping. However, as recent work emphasizes, theoretical incentive compatibility does not guarantee behavioral incentive compatibility in practice.

Complexity and behavioral incentive compatibility A major recent development in belief elicitation is the recognition that complex incentive schemes may fail because respondents do not fully understand them. [Abeler et al. \(2025\)](#) show experimentally that incentive complexity can reduce effort provision. Their model formalizes complexity as a cognitive cost that distorts optimization.

[Danz et al. \(2022\)](#) and [Danz et al. \(2024\)](#) extend this insight to belief elicitation. They demonstrate that several widely used mechanisms, such as quadratic and binarized scoring rules, often lead to systematic deviations in the reporting of an induced prior. They propose behavioral incentive compatibility as a robustness criterion that accounts for limited understanding, heuristics, and cognitive costs.

This perspective naturally connects to the survey methodology literature. [Krosnick \(1991\)](#) shows that when questions impose high cognitive burden, respondents adapt simplifying strategies, such as rounding and focal-point answers. [Smith and Walker \(1993\)](#) similarly argue that monetary incentives interact with task complexity in shaping decision costs. These findings imply that incentive schemes with complicated payoff structures may be dominated, in practice, by simpler schemes even if the latter lack theoretical incentive compatibility.

Our study bridges the macroeconomic expectations literature with the belief elicitation literature by experimentally comparing incentive schemes that differ in both their theoretical properties and their behavioral complexity. [Drobot et al. \(2024\)](#) previously examined incentivized inflation expectations, but existing work has not systematically contrasted mechanisms with varying levels of theoretical incentive compatibility and complexity. Similarly, although behavioral incentive compatibility has become central in experimental economics ([Danz et al. 2022](#); [Danz et al. 2024](#); [Abeler et al. 2025](#)), its implications for the elicitation of macroeconomic expectations remain unexplored.

By varying both the theoretical incentive compatibility and the complexity of the elicitation mechanism, our experiment directly tests whether more theoretically appealing schemes produce better behavioral performance—or whether the cognitive demands of these mechanisms undermine their effectiveness relative to simpler, more intuitive alternatives. In doing so, our study provides both methodological guidance for surveys of macroeconomic expectations and broader evidence on how complexity interacts with incentives in shaping reported beliefs.

2 Theoretical Framework

This section develops a unified framework for the two elicitation tasks used in our experiment and in the New York Fed’s Survey of Consumer Expectations (SCE):

1. Inflation point predictions;
2. Subjective probability distributions over future inflation (density forecast).

We first introduce common notation and then discuss the incentive properties and complexity trade-offs of the scoring rules used in each treatment. We leave the formal proofs to [Appendix A3](#).

2.1 Common Environment and Notation

Let $\pi \in \mathbb{R}$ denote the realized rate of inflation over the next 12 months. Each respondent holds a subjective belief over π , represented by a cumulative distribution function F with associated density f .

Point prediction task. In the point prediction task, the respondent reports a single real number $g \in \mathbb{R}$, interpreted as their point forecast of π .

Density prediction task. In the density prediction task, the support of π is partitioned into $K = 10$ disjoint intervals (bins),

$$\{b_k\}_{k=1}^K, \quad \bigcup_{k=1}^K b_k = \mathbb{R}, \quad b_k \cap b_j = \emptyset \text{ for } k \neq j.$$

The respondent reports a probability vector

$$\mathbf{g} = (g_1, \dots, g_K) \in \Delta^{K-1},$$

where g_k denotes the reported probability that $\pi \in b_k$, with $\sum_{k=1}^K g_k = 1$.

The respondent's true subjective probabilities implied by F are

$$p_k = \Pr(\pi \in b_k), \quad \mathbf{p} = (p_1, \dots, p_K) \in \Delta^{K-1},$$

where Δ^{K-1} denotes the $(K - 1)$ -dimensional probability simplex.

Scoring rules. Let $S_P(g, \pi)$ denote the payoff-relevant scoring rule for point forecasts and $S_D(\mathbf{g}, \pi)$ the scoring rule for density forecasts. Unless otherwise stated, incentive-compatibility results in this section are derived under risk neutrality. When risk preferences matter, we state this explicitly.

Complexity. Beyond incentive compatibility, we emphasize the *complexity* of the scoring rule, which we interpret as difficult it is for respondents to understand the mapping between their reports and resulting payoffs. As emphasized by [Charness et al. \(2021\)](#) and [Danz et al. \(2022\)](#), complexity can attenuate the behavioral effectiveness of otherwise theoretically clean incentives. Further, [Abeler et al. \(2025\)](#) demonstrate carefully using both laboratory and field experiments that complex incentives can distort effort relative to a rational benchmark.

2.2 Point Predictions

Laboratory and survey experiments that elicit point predictions for macroeconomic variables frequently adopt incentive schemes based on the learning-to-forecast methodology introduced by [Marimon et al. \(1993\)](#). Typically, such experiments use payment rules with quadratic loss functions or absolute value loss functions. Some recent examples include [Colasante et al. \(2020\)](#), [Evans et al. \(2021\)](#) [Xiao \(2024\)](#), and [Alfarano et al. \(2023\)](#), with quadratic loss functions, and [Rybacki \(2020\)](#), [Rholes and Petersen \(2021\)](#), [Mokhtarzadeh and Petersen \(2021\)](#), [Petersen and Rholes \(2022\)](#), and [McMahon and Rholes \(2023\)](#) with absolute value loss functions.

Under risk neutrality, minimizing expected quadratic loss elicits the mean of the belief distribution F , while minimizing expected absolute loss elicits the median of F . In this paper, we implement an exponential scoring rule that is a monotone transformation of the absolute loss:

$$S_P(g, \pi) = a \times 2^{-|g - \pi|}, \tag{1}$$

where $a > 0$ is a scaling constant set to $a = 10$ in our experiment, to avoid negative payoffs.¹

Treatment *Both*. In our simplest incentivized treatment, *Both*, respondents are told that their payoff declines sharply as their forecast error increases. They see only verbal explanations and numerical examples (e.g., that the payment halves when the point forecast error rises by one percentage point in absolute terms). Importantly, we do not reveal the explicit objective function, (Equation (1)). Nevertheless, the rule strongly rewards accuracy, is incentive-compatible under risk neutrality and transparent, making it a low-complexity mechanism that is likely to increase cognitive effort.

Treatment *Both+Formula*. In *Both+Formula*, respondents receive all information available to respondents in *Both*, but also learn the explicit scoring function (Equation (1)). Under risk neutrality, the respondent chooses a report g to maximize expected payoff

$$\mathbb{E}_F[S_P(g, \pi)] = \int S_P(g, \pi) dF(\pi).$$

For the exponential scoring rule, this objective is equivalent to minimizing

$$\int |g - \pi| dF(\pi),$$

which is uniquely minimized at the median of the subjective distribution. Hence, any optimal report g^* satisfies

$$\Pr(\pi \leq g^*) = \frac{1}{2}.$$

Relative to *Both*, on the one hand, this treatment is theoretically cleaner as we provide a functional form for the loss function that may make incentives more transparent. On the other hand, it is possibly more complex for respondents due to the explicit inclusion of the mathematical representation of incentives.

¹For other examples using this scoring rule, see [Drobot et al. \(2024\)](#), [McMahon and Rholes \(2023\)](#), [Rholes and Petersen \(2021\)](#), [Mokhtarzadeh and Petersen \(2021\)](#). Here, $a = 10$ implies a perfect forecast where $g = \pi$ earns a respondent a \$10 bonus payment.

Treatment *Median* (binarized scoring). In the *Median* treatment, we binarize the scoring rule by interpreting (Equation (1)), with $a = 100$, as the probability of receiving a fixed \$10 bonus payment. Under the standard assumptions that participants maximize expected utility and are neutral to the randomization device, binarization eliminates distortions from risk preferences and preserves median elicitation for all money-motivated agents (Hossain and Okui 2013).² We therefore view *Median* as the most theoretically disciplined design for point forecasts. The cost is higher complexity: subjects must understand both a nonlinear accuracy mapping (i.e., *Both +Formula*) that produces a lottery over payments.

2.3 Subjective Probability Forecasts

There are several established methods for eliciting subjective probability distributions (Eytting and Schmidt 2021). One approach is the elicitation of interval probabilities, which divides the outcome space into mutually exclusive bins and asks respondents to allocate probability mass across them, subject to summing to one. Another approach elicits multiple quantile forecasts, requiring respondents to report specific percentiles of their subjective distribution. More recently, Goldfayn-Frank et al. (2025) propose a choice-based elicitation method.

Because our experimental design mirrors the format of the New York Fed’s *Survey of Consumer Expectations* (SCE), we elicit subjective probabilistic forecasts using interval probabilities. Respondents therefore allocate probability mass across the ten inflation bins $\{b_k\}_{k=1}^{10}$ defined in the survey instrument.

Treatments *Both* and *Both+Formula*: Simple rule. In our simplest incentivized treatments, *Both* and *Both+Formula*, we adopt the same “pay-the-probability” scoring rule used by Drobot et al. (2024), chosen for its extreme simplicity and transparency. Specifically,

²Recent work by Danz et al. (2022) shows that binarized scoring rules are not necessarily incentive compatible under ambiguity aversion. If respondents are ambiguity averse, they may distort reports to smooth the probability of earning the bonus across states. This concern is potentially more salient for probability forecasts, where respondents can mechanically guarantee a fixed winning probability by reporting uniform beliefs. There is currently no general solution to this issue; we follow the standard practice in the literature by abstracting from ambiguity preferences.

if realized inflation falls into bin b_k , the respondent earns

$$S_D^{\text{simple}}(\mathbf{g}, \pi) = 10 \times g_k \quad \text{if } \pi \in b_k. \quad (2)$$

In *Both*, respondents receive only verbal explanations and numerical examples. In *Both+Formula*, the explicit mathematical form of the rule is also shown.

Under risk neutrality, this rule is *not* incentive compatible for truthful reporting of $\mathbf{p}$: expected payoff is linear in $\mathbf{g}$, implying that any report placing all mass on a modal bin maximizes expected earnings (see [Appendix A3](#)). Thus, the rule does not recover the full subjective distribution in theory under risk neutrality. Its appeal lies instead in its extreme simplicity and transparency.

Our key theoretical contribution is to show that the properties of this rule change fundamentally under risk aversion. We assume constant-relative-risk-aversion (CRRA) preferences over monetary payoffs,

$$U(x) = \begin{cases} \frac{x^{1-\rho}}{1-\rho}, & \rho \neq 1, \\ \ln x, & \rho = 1, \end{cases} \quad \rho \geq 0,$$

and characterize the optimal report under (Equation (2)), which ultimately yields $g_k^* \propto p_k^{1/\rho}$.

Proposition 1 (CRRA distortion under $S_D^{\text{simple}}(\mathbf{g}, \pi)$). *Let $\mathbf{p} = (p_1, \dots, p_K) \in \Delta^{K-1}$ denote the respondent's true beliefs over bins $\{b_k\}_{k=1}^K$, and suppose preferences are CRRA with coefficient $\rho > 0$. Under the simple pay-the-probability rule (Equation (2)), any interior optimum $\mathbf{g}^*(\rho)$ that maximizes expected utility satisfies*

$$g_k^*(\rho) = \frac{p_k^{1/\rho}}{\sum_{j=1}^K p_j^{1/\rho}}, \quad k = 1, \dots, K. \quad (3)$$

Moreover:

1. As $\rho \rightarrow 0^+$ (risk neutrality), $\mathbf{g}^*(\rho)$ concentrates all mass on the modal set $M = \{k : p_k = \max_j p_j\}$.

2. At $\rho = 1$ (log utility), reports are exactly truthful: $g_k^*(1) = p_k$ for all k .
3. As $\rho \rightarrow \infty$ (extreme risk aversion), reports converge to the uniform distribution: $g_k^*(\rho) \rightarrow 1/K$ for all k .

An immediate implication of the structure of (Equation (3)) is that the distortion induced by risk aversion is reversible. For any fixed $\rho > 0$, the mapping from true beliefs to reported probabilities is bijective on the probability simplex and therefore admits a unique inverse.

Corollary 1 (Invertibility). *Fix $\rho > 0$. The mapping $\mathbf{p} \mapsto \mathbf{g}^*(\rho)$ in (Equation (3)) is bijective on the probability simplex. In particular, for any report $\mathbf{g}$ generated by CRRA preferences with coefficient ρ , the latent beliefs are recovered by*

$$p_k = \frac{g_k^\rho}{\sum_{j=1}^K g_j^\rho}, \quad k = 1, \dots, K. \quad (4)$$

We provide proofs of proposition 1 and corollary 1, including a detailed characterization of the induced total-variation distortion between $\mathbf{p}$ and $\mathbf{g}^*(\rho)$, in [subsection A3.4](#).

These results deliver two key implications. First, the simple pay-the-probability rule is not merely a degenerate heuristic under risk aversion, as it is under risk neutrality. Instead, it behaves as a well-structured distortion of a proper rule that is exactly truthful at $\rho = 1$, and varies continuously between modal point-mass reports ($\rho \rightarrow 0^+$) and a uniform distribution ($\rho \rightarrow \infty$). Second, because the distortion admits the closed form in Equation (3) and is invertible via Equation (4), its magnitude can be quantified and, under assumptions on ρ , partially corrected. These properties are central to our interpretation of why the simple rule can outperform more complex mechanisms in practice, despite its lack of formal incentive compatibility under risk neutrality. The recoverability result also partially motivates our inclusion of an incentivized risk elicitation task, described in detail in the experimental design.

Treatment *Median*: Quadratic scoring rule with binarization. In our theoretically disciplined treatment, *Median*, we adopt the most widely used approach for incentivizing the elicitation of interval probabilities – a generalized version of the quadratic scoring rule (QSR), which is theoretically incentive compatible under risk neutrality (Danz et al. 2022; Eytting and Schmidt 2021; Palley and Bansal 2021; Harrison et al. 2015):

$$S_D^{\text{QSR}}(\mathbf{g}, \pi) = a + bg_k - a \sum_{i=1}^{10} g_i^2, \quad (5)$$

with $b = 2a > 0$. Under risk neutrality, this rule is strictly proper: the unique report that maximizes expected payoff is $\mathbf{g} = \mathbf{p}$. See Appendix A3, subsection A3.3 for proof.

As with $S_P(g, \pi)$ in *Median*, we implement (Equation (5)) in binarized form by mapping the score into the probability of earning a \$10 bonus. This removes the influence of risk preferences and yields incentive compatibility for density forecasts under broad conditions. As before, the trade-off is increased cognitive complexity due to both the quadratic structure and the lottery over payoffs.

2.4 Summary of Incentive Properties and Testable Implications

Our four treatments therefore differ along two dimensions: (i) theoretical incentive compatibility and (ii) complexity.

- **Control:** No performance-based incentives.
- **Both:** Simple, example-based incentives for both point and density forecasts. Low complexity, but not formally incentive compatible.
- **Both+Formula:** Same payoffs as *Both*, with explicit formulas. Theoretically incentive compatible for point forecasts under risk neutrality, but higher complexity.
- **Median:** Binarized exponential scoring for point forecasts and binarized quadratic scoring for density forecasts. Fully incentive compatible under broad conditions, but

the most complex.

This structure generates two central trade-offs that guide our empirical analysis. First, introducing even simple incentives should increase effort and reduce noise relative to the *Control* group. Second, while *Median* is theoretically optimal, its higher complexity may attenuate its behavioral effectiveness relative to the simpler *Both* treatment.

3 Survey Design

Our survey replicates the inflation expectations questions from the New York Fed’s SCE, enabling direct comparability with a benchmark dataset widely used by researchers and policymakers.

To identify the effect of marginal incentives on belief formation, we randomly assigned respondents to one of four groups that differed only in whether and how forecast accuracy affected their bonus payment. All respondents received a flat participation fee and an additional bonus payment one year later. For respondents in the *Control*, this bonus was fixed and known at the outset of the experiment and did not depend on forecast performance. This directly replicates the flat-fee incentive structure used in the SCE.

In *Median*, *Both*, and *Both+formula*, bonus payments depended directly on the accuracy of respondents’ one-year-ahead inflation forecasts, thereby introducing marginal incentives into the expectations-formation task. We informed respondents in these treatments that we would randomly select either their point or density forecast of inflation for payment with equal probability. In *Median*, we elicited point forecasts of inflation expectations using the binarized version of $S_P(g, \pi)$ described in Section 2.2 and density forecasts with $S_D^{\text{QSR}}(\mathbf{g}, \pi)$, described in Section 2.3. In *Both* and *Both+formula*, we elicited point forecasts using the non-binarized version of $S_P(g, \pi)$ and density forecasts with $S_D^{\text{simple}}(\mathbf{g}, \pi)$. Relative to *Both*, *Both+formula* explicitly displayed these scoring rules, allowing us to test whether instructional complexity effects forecast behavior. Table 1 summarizes our treatments.

Group	Point Forecast Bonus Rule	Density Forecast Bonus Rule
<i>Flat Incentives</i>		
Control	Bonus = \$4	Bonus = \$4
<i>Marginal Incentives</i>		
Both	Bonus = $\$10 \times 2^{- \text{error} }$	Bonus = $\$10 \times P$
Both+Formula	Bonus = $\$10 \times 2^{- \text{error} }$	Bonus = $\$10 \times P$
Median	$\Pr(\text{Bonus} = \$10) = 100 \cdot 2^{- \text{error} }$	$\Pr(\text{Bonus} = \$10) = 50 + 100p_k - 50 \sum_{i=1}^{10} p_i^2$

Table 1: Scoring Rules by Experimental Group

Notes: We randomly selected either the point or density forecast for payment in all incentivized treatments. In the *Both* treatment, we use examples and verbal descriptions to show how forecast accuracy maps into payment. In the *Both+formula* treatment, we add the explicit scoring-rule formulas to these examples. In the *Median* treatment, we present the same information and also implement binarized scoring rules, which raise the instructional complexity by introducing probability-based payments and the associated strict-properness logic.

After eliciting one-year-ahead inflation forecasts, we collect unincentivized three-year-ahead inflation forecasts to examine potential spillover effects. We then reminded respondents that subsequent questions would not affect their bonus payments and asked follow-up questions about their forecasts, including an open-ended question, to gain deeper insights into the elicited beliefs and the expectation formation process more broadly. We also asked whether respondents consulted any information sources while forming their forecasts, which allows us to test whether incentives encourage information search.

Next, we measured respondents’ risk preferences using the Bomb Risk Elicitation Task (BRET), which enables us recover true underlying belief distributions to via Equation (3).³ We then elicit additional expectations related to income, spending, and prices for various categories of goods and services, including gasoline, food, medical care, education, and rent. Finally, assess cognitive ability and numerical literacy, and conclude the survey by collecting standard demographic information.

³See [Crosetto and Filippin \(2013\)](#) and [Charness et al. \(2013\)](#).

3.1 Survey Data

We conducted the online survey in March 2025 using the Prolific platform. In total, we collected 1,024 responses, with approximately equal numbers assigned to each treatment group. The sample was drawn from a nationally representative adult population. [Table A-1](#) shows that the demographic composition is relatively balanced across the treatment groups.

4 Results

4.1 Effect of Incentives on Point Forecasts

In line with our previous findings ([Drobot et al. 2024](#)) and consistent with established results in the experimental economics literature ([Smith and Walker 1993](#)), we find that incentives have a strong effect on respondents’ point forecasts. [Figure 1](#) illustrates that the treatments noticeably reshape the distribution of elicited inflation expectations, as measured by the point forecast question. Specifically, the unincentivized group shows a more dispersed set of responses and exhibits a noticeable “bump” around +50% inflation. This pattern may indicate that some respondents did not pay close attention to the question and interpreted it as asking about the general likelihood of inflation occurring, rather than providing a precise numerical forecast.

4.1.1 Effort

We find that respondents in the incentivized groups spend more time answering the point forecast question compared to those in the unincentivized (*Control*) group, suggesting higher cognitive effort (see [Figure A-1](#)). Notably, the increase in response time appears to reflect deeper consideration rather than a greater tendency to seek external information. As shown in [Figure A-2](#), most respondents do not search for inflation-related information online when forming their forecasts, and the introduction of incentives leads to only a modest increase in

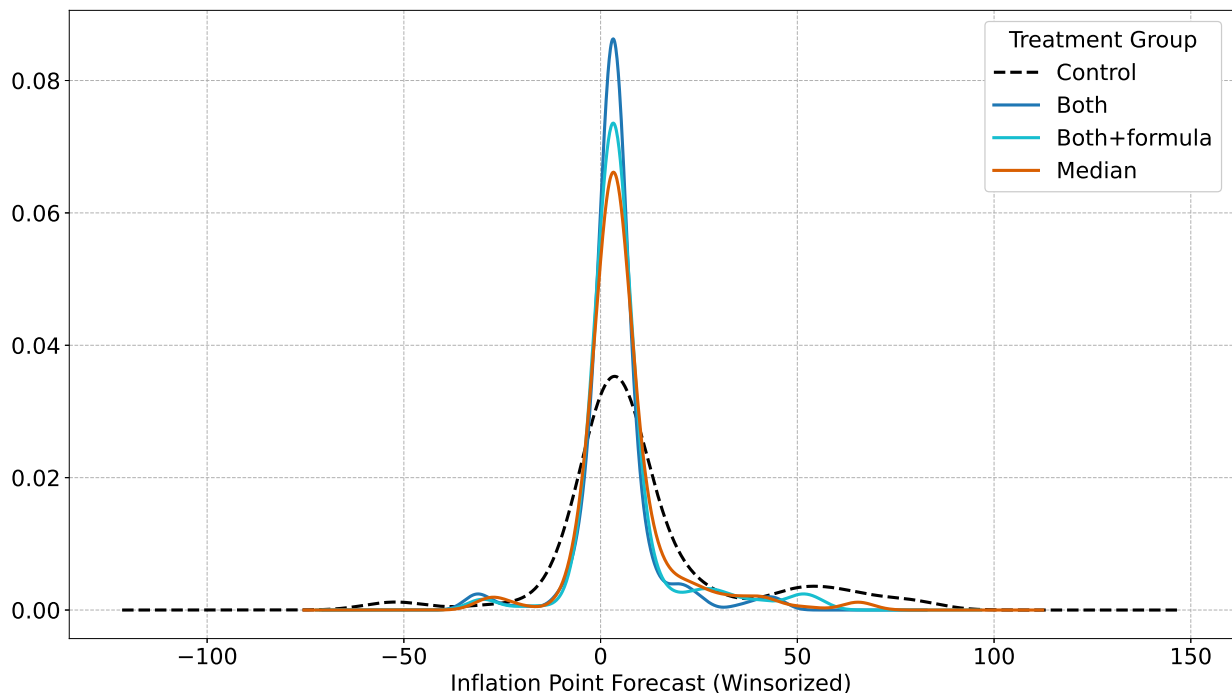


Figure 1: PDF of Inflation Point Forecast

Notes: The figure shows the kernel density estimation of winsorized inflation point forecasts by treatment group. Winsorization is applied at the 1st and 99th percentiles to limit the influence of extreme values.

such search behavior. Furthermore, this result is not driven by the additional effort required to read the instructions, as our survey is designed so that the point forecast question and the instructions appear on separate screens, allowing us to track the time spent on each task separately.

4.1.2 Extreme Forecasts

In this section, we examine the effect of incentives on extreme forecasts, motivated by evidence that extreme responses can reflect either misunderstanding of the question (De Bruin et al. 2012, Armantier et al. 2013) or satisficing behavior (Krosnick 1991), both of which suggest deviations from truthful reporting.

We define an extreme forecast dummy variable (F_e) as taking the value of 1 if the absolute value of the point forecast falls at or above the 90th percentile, capturing both extremely high inflation and extreme deflation forecasts.

To quantify the effect of incentives on the likelihood of reporting an extreme forecast, we estimate a logistic regression, with results presented in [Table 2](#), Columns 1 and 2. As shown, the probability of providing an extreme forecast is significantly lower among respondents in the incentivized groups relative to the control group.

	Dependent variable: F_e		Dependent variable: $\mathbb{E}[\pi_i]$	
	(1)	(2)	(3)	(4)
<i>Both</i>	-1.215*** (0.304)	-1.301*** (0.311)	-6.242*** (1.292)	-6.215*** (1.293)
<i>Both + formula</i>	-1.357*** (0.319)	-1.480*** (0.326)	-4.313*** (1.297)	-4.306** (1.297)
<i>Median</i>	-1.267*** (0.311)	-1.342*** (0.318)	-4.185*** (1.297)	-4.101*** (1.297)
Constant	-1.497*** (0.161)	-2.141*** (0.385)	10.004*** (0.914)	7.148*** (1.644)
Controls	No	Yes	No	Yes
Regression	Logit	Logit	OLS	OLS
N	1,024	1,024	1,024	1,024

Table 2: Extreme Forecast Reporting and Upward Bias

Notes: Columns (1) and (2) present the results of logistic regressions where the dependent variable is the extreme forecast dummy variable F_e . Columns (3) and (4) report results from OLS regressions where the dependent variable is the point forecast $\mathbb{E}[\pi_i]$. The data are winsorized at the 1st and 99th percentiles to reduce the influence of extreme values. Models in columns (2) and (4) include controls for demographic characteristics, including age, gender, education, employment, and race. * $p < .1$, ** $p < .05$, *** $p < .01$

4.1.3 Biases

The economic literature has extensively documented several systematic biases in household inflation expectations. One of the most persistent is the *upward bias*, whereby households tend to forecast inflation rates that exceed both realized outcomes and the forecasts produced by professional forecasters. [D’Acunto et al. \(2023\)](#) attribute this bias to a behavioral tendency: households overweight price increases relative to price decreases when forming expectations.

In our setting, we propose a complementary mechanism: the lack of incentives in traditional survey settings may discourage respondents from carefully considering the future, particularly in ways that mimic real-world financial decision-making. If incentives prompt respondents to engage more thoughtfully with the forecasting task, we would expect them to report beliefs that more closely reflect the expectations they would hold in real-life contexts — such as when making important financial or consumption decisions — rather than providing overly simplified or heuristic-based responses that may contribute to upward bias.

We test this hypothesis by estimating the following linear regression:

$$\mathbb{E}[\pi_i] = \beta_0 + \sum_{k=1}^3 \beta_k \cdot \text{Treatment}_{k,i} + \mathbf{W}_i' \boldsymbol{\phi} + \varepsilon_i \quad (6)$$

where $\mathbb{E}[\pi_i]$ is the one-year-ahead inflation forecast of respondent i , $\text{Treatment}_{k,i}$ is a dummy variable indicating exposure to treatment $k \in \{\text{Both}, \text{Both+Formula}, \text{Median}\}$, and $\mathbf{W}_i$ is a vector of demographic controls including gender, age, education, employment status, and race. The reference category is the *Control* group, which did not receive any incentive.

As shown in [Table 2](#), all three treatments significantly reduce inflation forecasts relative to the control group. Notably, the effect is strongest for the simplest incentive (*Both*), suggesting that even straightforward, easy-to-understand incentive schemes can prompt respondents to engage more carefully with the task and provide more thoughtful and well-considered expectations.

Following [Carroll \(2001\)](#), we use the Survey of Professional Forecasters (SPF) as a benchmark for “rational forecast” that respondents can learn from the media. We find that incentivized groups report aggregate forecasts that are closer to those of professionals, indicating a reduction in upward bias and an overall shift toward “rationality” (see [Table A-2](#)). Moreover, we observe a decline in the cross-sectional dispersion of forecasts among incentivized respondents ([Table A-3](#)), suggesting that incentives promote not only greater alignment with professional forecasts — who are typically more informed — but also convergence in beliefs,

consistent with findings from the experimental economics literature.

We further investigate the effect of incentives on well-documented expectation gaps across gender, age, income, political affiliation, and cognitive ability (D'Acunto et al. 2023, Pfajfar and Santoro 2008). Figure 2 presents the results. Across most dimensions, we document a meaningful reduction in these gaps when incentives are introduced. While some subgroup differences remain statistically insignificant (potentially due to macroeconomic conditions beyond survey design) the pattern is clear: even the simplest incentive (*Both*) consistently narrows these gaps. This finding reinforces the importance of accounting for respondent motivation in survey design and highlights incentives as a practical tool to improve the reliability and reproducibility of survey-based economic research, making it less dependent on external conditions.

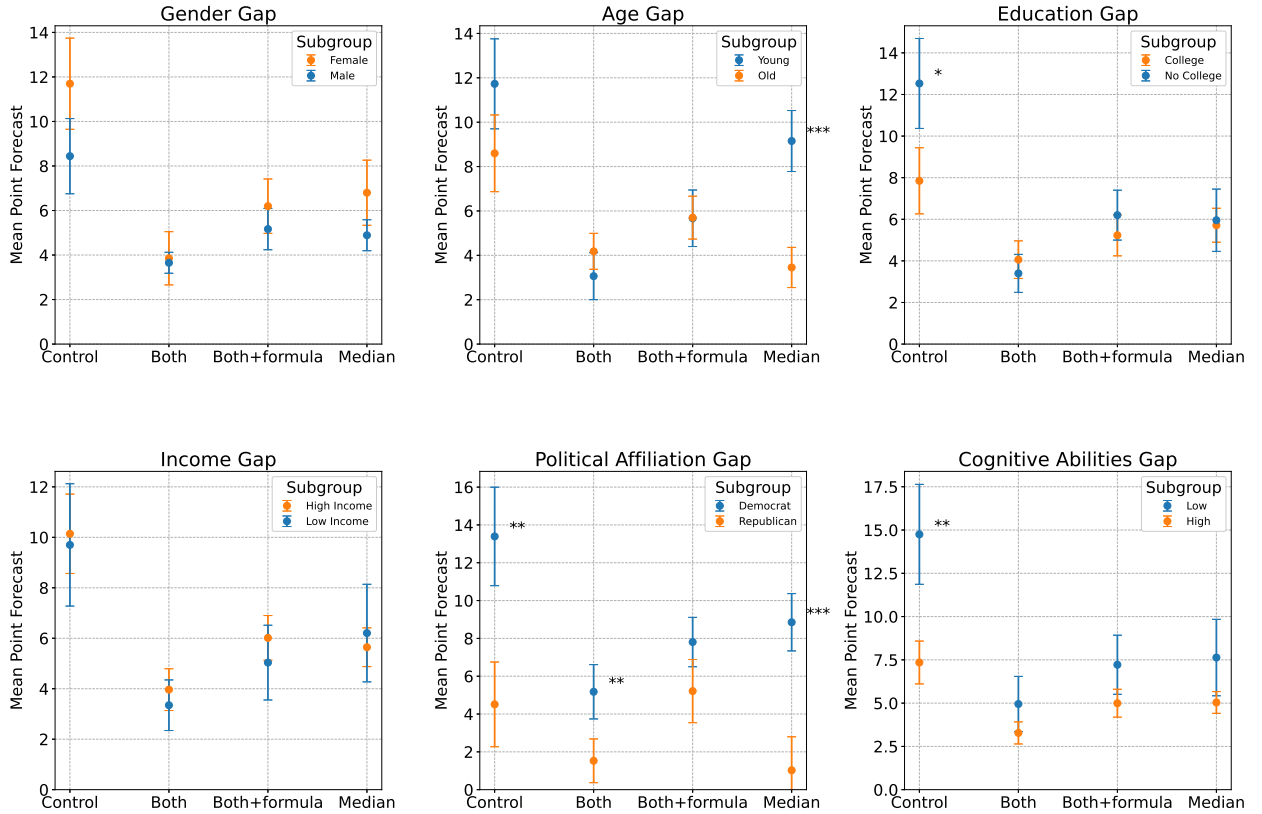


Figure 2: Inflation Expectations Gaps

Notes: The figure shows the effect of incentives on various inflation expectation gaps. Markers represent the average forecast for each subgroup, with vertical bars indicating $\pm$ one standard error of the mean. Stars denote statistically significant differences between subgroups within each treatment based on Welch's two-sample t -tests: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

4.1.4 Spillover Effects

So far, we see that incentives have a strong impact on one-year-ahead inflation point forecasts. But do they also affect other, non-incentivized parts of the survey? After eliciting one-year-ahead inflation forecasts, we also ask respondents about three-year-ahead inflation expectations, perceived current inflation, and expectations for twelve-month-ahead price growth in specific categories: gasoline, food, medical care, education, and rent. These questions are clearly marked as non-incentivized.

In [Table A-4](#), we document a spillover effect on medium-term inflation forecasts — a reduction in the mean forecast value relative to the control group. This supports the interpretation that the treatment induced a cognitive shift in expectations, rather than a mere strategic adjustment to comply with the experimental setting. Consistent with our previous findings ([Drobot et al. 2024](#)), we also observe that respondents’ reported perceived inflation is, on average, much closer to the true value (see [Figure 3](#)). Importantly, this result does not appear to be driven by increased information search. As noted earlier, incentivized respondents do not report noticeably higher search activity than the unincentivized group, suggesting that the improvement is more likely due to greater cognitive effort and a reduction in emotional biases ([Figure A-2](#)).

However, we do not observe a corresponding effect of incentives on forecasts for individual components of inflation. Given prior research showing that households tend to overweight salient categories, such as food and gasoline, when forming inflation expectations (e.g., [Cav-allo et al. 2017](#); [D’Acunto et al. 2021](#)), our findings may indicate that incentives encouraged respondents to rely less on these salient components and instead anchor their overall forecast on more realistic, aggregate-consistent weights.

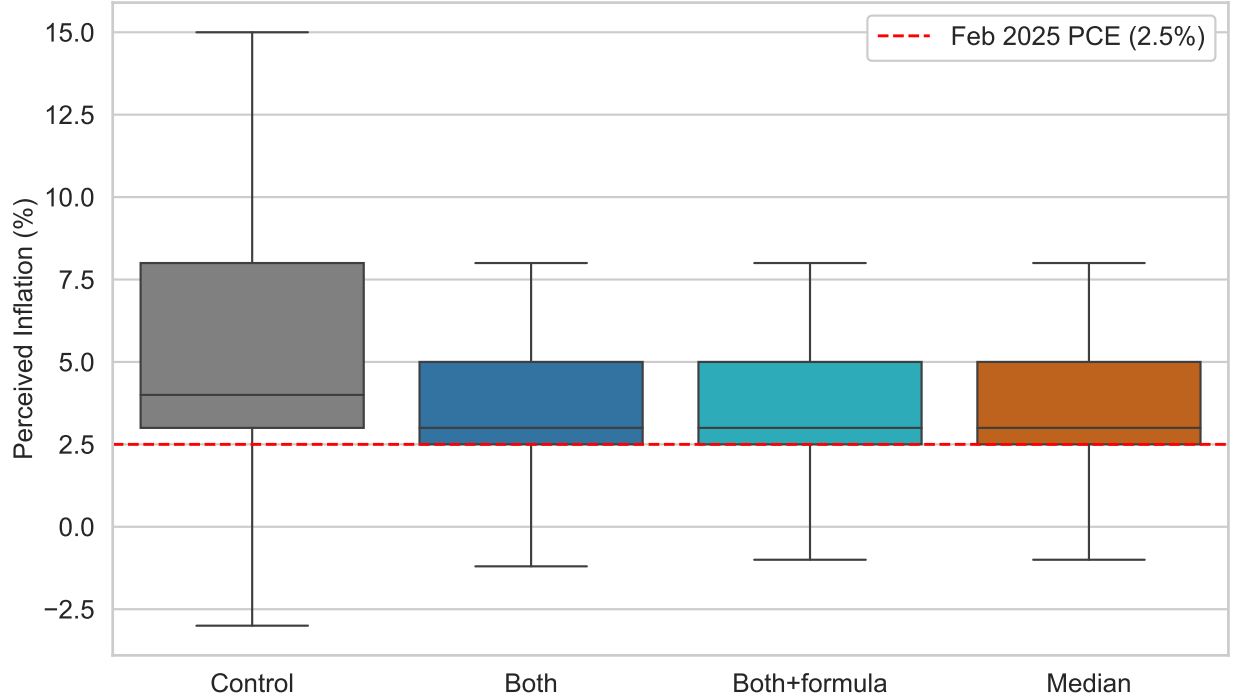


Figure 3: Perceived Inflation

Notes: The figure presents a boxplot of perceived inflation across different treatment groups. The red dashed horizontal line indicates the most recently published level of PCE inflation at the time of the survey.

4.1.5 Backward-Looking Heuristics

The link between perceived and expected inflation is well documented in the literature [Jonung \(1981\)](#). We quantify the strength of this relationship using the following regression:

$$\mathbb{E}[\pi_i] = \alpha_0 + \alpha_1 P_i + \mathbf{W}_i' \boldsymbol{\phi} + \varepsilon_i, \quad (7)$$

where P_i denotes individual perceived inflation. We report α_1 , the coefficient that captures the strength of the association between expected and perceived inflation, in Figure 4. The results are striking: incentives, particularly under the least comprehensive scoring rule, weaken the link between the two variables, reducing individuals' reliance on backward-looking heuristics.

To better understand the intuition behind these findings, we turn to an open-ended question in which respondents were asked to explain their forecasts. We find no substantial differences

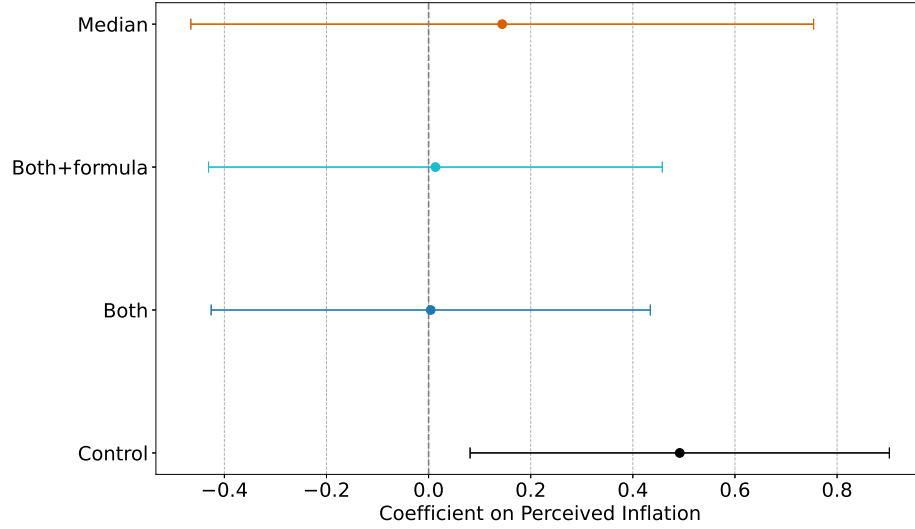


Figure 4: Robust OLS Coefficient on Perceived Inflation by Treatment

Notes: The figure displays the estimated coefficient on perceived inflation from separate OLS regressions for each treatment group. The dependent variable is the respondent’s point forecast of inflation, winsorized at the 1th and 99th percentiles. The key independent variable is perceived inflation (also winsorized), and all regressions control for gender, education level, age category, employment status, and race. The plotted points represent point estimates, and the horizontal lines indicate 99% robust confidence intervals. A dashed vertical line at zero provides a visual benchmark for statistical significance.

across groups in either the length of the responses or the logic and factors cited to justify expectations. Our simple measure of backward- and forward-looking orientation in text responses detects only a slightly higher share of backward-looking responses in the *Control* group. However, the difference is small. Interestingly, the *Both* group shows the largest share of backward-looking responses across incentivized groups while simultaneously exhibiting the weakest link between perceived and expected inflation. [Appendix A2](#) provides details on this textual analysis. The LLM-based analysis also does not detect noticeable differences between groups. However, compared to our keyword-based classification, the *Both* group shows the highest share of forward-looking responses, while the other incentivized groups exhibit slightly lower forward-looking responses relative to the unincentivized group.

Taken together, these findings suggest that incentives did not alter the underlying mental models respondents used to form their forecasts — they merely attenuated the influence of perceived inflation in shaping those expectations.

4.2 Effect of Incentives on Density Forecasts

Eliciting inflation expectations through density forecast questions allows us to examine how incentives shape the individuals’ forecast distributions. Figure 5 presents the average weight that respondents assign to each bin. Notably, even in the absence of incentives, respondents place the highest weight on the “2 to 4” percent bin. This is particularly interesting, as this range aligns with an informed benchmark capturing both the Federal Reserve’s long-term inflation target and the mean forecast from the SPF, as well as the recent level of realized inflation.

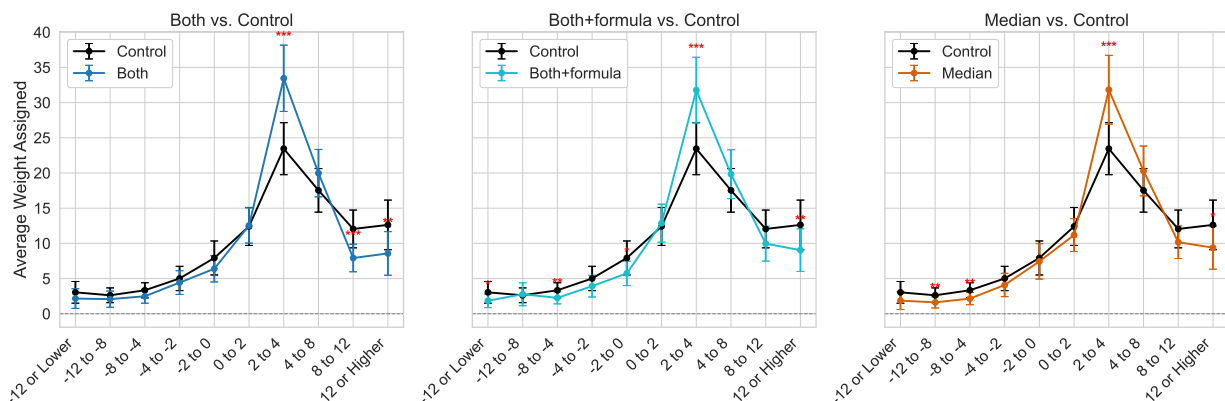


Figure 5: Average Bin Weights: Treatment vs. Control

Notes: The figure presents average weights assigned to inflation forecast bins by incentivized group, compared to the Control group. Each panel displays the estimated mean bin weights along with 99% confidence intervals. Welch’s two-sample t -test is used to test for significant differences between each treatment group and the Control group. Significance stars (*, **, ***) indicate statistically significant differences from the Control group at the 10%, 5%, and 1% levels, respectively.

Introducing incentives significantly increases the weight respondents assign to this central bin, while simultaneously reducing the weight allocated to the most extreme bin (“12 or Higher”). This pattern suggests that incentives help anchor expectations closer to conventional economic benchmarks and diminish perceived tail risks. This observation provides important food for thought for the future of monetary policy that aims to anchor inflation expectations around the target. It underscores the importance not only of delivering clear and accessible policy communication that reduces the costs of processing information, but also of making the underlying incentives for accurate forecasting more salient to households. By stimulating awareness of the personal and economic value of forming well-informed in-

flation expectations, policymakers may enhance the alignment between individual behavior and macroeconomic objectives.

The density elicitation method also enables a richer characterization of beliefs, as it allows us to estimate not only measures of central tendency (mean and median), but also higher-order moments such as variance and skewness. These moments offer insights into individual-level uncertainty as well as potential directional bias, such as optimism or pessimism. To quantify these characteristics, we follow the methodology used in the SCE by [Armantier et al. \(2017\)](#), which builds on [Engelberg et al. \(2009\)](#). Specifically, we fit a Generalized Beta distribution to each respondent’s density forecast.

To assess the effect of incentives on these distributional features, we estimate the following regression:

$$Y_i = \gamma_0 + \sum_{k=1}^3 \gamma_k \cdot \text{Treatment}_{k,i} + \mathbf{W}_i' \boldsymbol{\phi} + \varepsilon_i \quad (8)$$

where Y_i represents the respondent-level mean, median, variance, or skewness of the fitted distribution. The coefficients γ_k capture whether these moments on average statistically differ from those in the baseline (*Control*) group. Table 3 presents the regression results. We find that incentives significantly affect respondents’ uncertainty, as measured by the variance, while there is no statistically significant effect on the central moments (mean or median).

It is worth noting that the SCE elicitation methodology lowers reported central tendencies, which could partly explain the absence of the effect of incentives on mean or median. In this regard, alternative approaches — such as the method proposed by [Goldfayn-Frank et al. \(2024\)](#) — may yield different outcomes. Nevertheless, for the purposes of this study, we deliberately adopted the SCE instrument, given its widespread use in the literature and relevance for comparability.

	Mean	Median	Variance	Skewness
Intercept	3.3767** (0.5235)	3.3575*** (0.5417)	13.3107*** (4.6158)	0.0859 (0.0527)
<i>Both</i>	-0.5157 (0.4118)	-0.5685 (0.4261)	-8.0920** (3.6333)	0.0556 (0.0414)
<i>Both+formula</i>	0.2604 (0.4145)	0.2787 (0.4290)	-5.8156 (3.6423)	-0.0109 (0.0416)
<i>Median</i>	0.1864 (0.4156)	0.2157 (0.4300)	-8.6016** (3.6634)	-0.0044 (0.0418)
Controls	Yes	Yes	Yes	Yes
<i>N</i>	932	932	999	999

Table 3: Effect of Incentives on Moments of Inflation Expectations

Notes: The table reports OLS estimates of the effects of treatments on the mean, median, variance, and skewness of respondents' inflation expectations. The *Control* group is the reference category. All regressions control for gender, age, education, employment status, and race. Standard errors are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$

4.3 Consistency

In this section, we examine the consistency of the survey responses. Consistency across different elicited beliefs is informative not only about data quality, but also about the extent to which respondents process information in a reasoned and economically coherent manner.

4.3.1 Nonparametric Analysis

Point Forecast and Density Forecast Weights

To assess consistency, we begin by analyzing the share of subjective probability that respondents assign to the density bin containing their stated point forecast. This metric allows us to distinguish whether the elicited point forecast represents a well-formed belief or a response driven by guesswork or incentive structures.⁴

Table 4 presents regression results showing the relative weight respondents in incentivized

⁴If respondents respond strategically, those in the *Median* group should allocate probability more uniformly across bins, since this guarantees a non-zero payment under the scoring rule if the question is selected.

	(1)	(2)
Intercept	36.204*** (1.888)	36.951*** (3.193)
<i>Both</i>	6.387** (2.484)	6.086** (2.478)
<i>Both+formula</i>	1.761 (2.511)	1.713 (2.537)
<i>Median</i>	6.642** (2.526)	6.558** (2.531)
Controls	No	Yes
<i>N</i>	1024	1024

Table 4: Effect of Incentives on Weights

Notes: The table presents the results of OLS regression where dependent variable is weight assigned to the bin containing the stated point forecast. Regression (2) controls for gender, age, education, employment status, and race. Standard errors are in parentheses. * $p < .1$, ** $p < .05$, *** $p < .01$

groups place on the bin that contains their point forecast, compared to the *Control* group. The results indicate that respondents in the *Both* and *Median* groups assign significantly higher weights to the stated bin, suggesting that incentives can enhance consistency between point forecast and density forecast.

Consistency of Point Predictions with the Bounds

Following Engelberg et al. (2009), we bound the mean, median, and mode of the density forecast question.⁵ We find that most respondents' point forecasts fall within the bounds of the mode (see Table 5), which aligns closely with respondents' own answers regarding the interpretation of their point forecasts (see Figure A-3).

4.3.2 Parametric Analysis

Mean or Median

⁵The only difference in our methodology is that we assume open intervals are bounded. We adopt this approach because household expectations data is much noisier compared to the SPF data used in Engelberg et al. (2009).

Group	# obs.	Share (%)			McNemar Test (p-value)		
		Median	Mean	Mode	Median vs Mean	Median vs Mode	Mean vs Mode
Control	257	36.6	47.5	49.4	0.000	0.000	0.615
Both	257	59.5	63.0	66.9	0.328	0.005	0.295
Both+formula	257	49.8	54.5	63.0	0.119	0.000	0.008
Median	253	51.4	56.9	64.8	0.087	0.000	0.022

Table 5: Upper bounds on mean/median/mode point predictions

Notes: The table reports share of respondents whose point forecast falls within mean/median/mode bounds of density forecast. McNemar tests compare whether the proportion of forecasts consistent with one measure of central tendency differs significantly from the proportion consistent with another measure, accounting for paired forecasts.

As discussed earlier, the absolute loss function theoretically elicits the median of a belief distribution. To test this implication, we compare the absolute distance between the point forecast and both the mean and the median of the corresponding density forecast. Because symmetric distributions trivially yield equal distances, we restrict the analysis to distributions exceeding certain skewness thresholds. We then test whether the average distance to the mean differs significantly from the average distance to the median.

Figure 6 presents the results. We find that only respondents in the *Median* group exhibit a consistent tendency for their point forecasts to align more closely with the median than with the mean, which supports theoretical results discussed in Part 2 of the paper.

Group	Absolute Distance To Mean				Absolute Distance To Median			
	Treatment	Control	T-test (p-value)	Mann-Whitney U (p-value)	Treatment	Control	T-test (p-value)	Mann-Whitney U (p-value)
Both	3.63	9.84	0.000	0.000	3.61	9.82	0.000	0.000
Both+formula	5.55	9.84	0.001	0.004	5.54	9.82	0.001	0.003
Median	4.47	9.84	0.000	0.002	4.39	9.82	0.000	0.000

Table 6: Comparison of Absolute Distances to Mean and Median

Notes: The table presents the mean absolute distances between respondents' point forecasts and the mean or median of their subjective density forecasts. The t-test (difference in means) and Mann-Whitney U test (difference in distributions) assess whether these distances differ significantly between the treatment groups and the control group.

In addition we compare internal consistency of our data to the SCE data, and find interesting pattern.

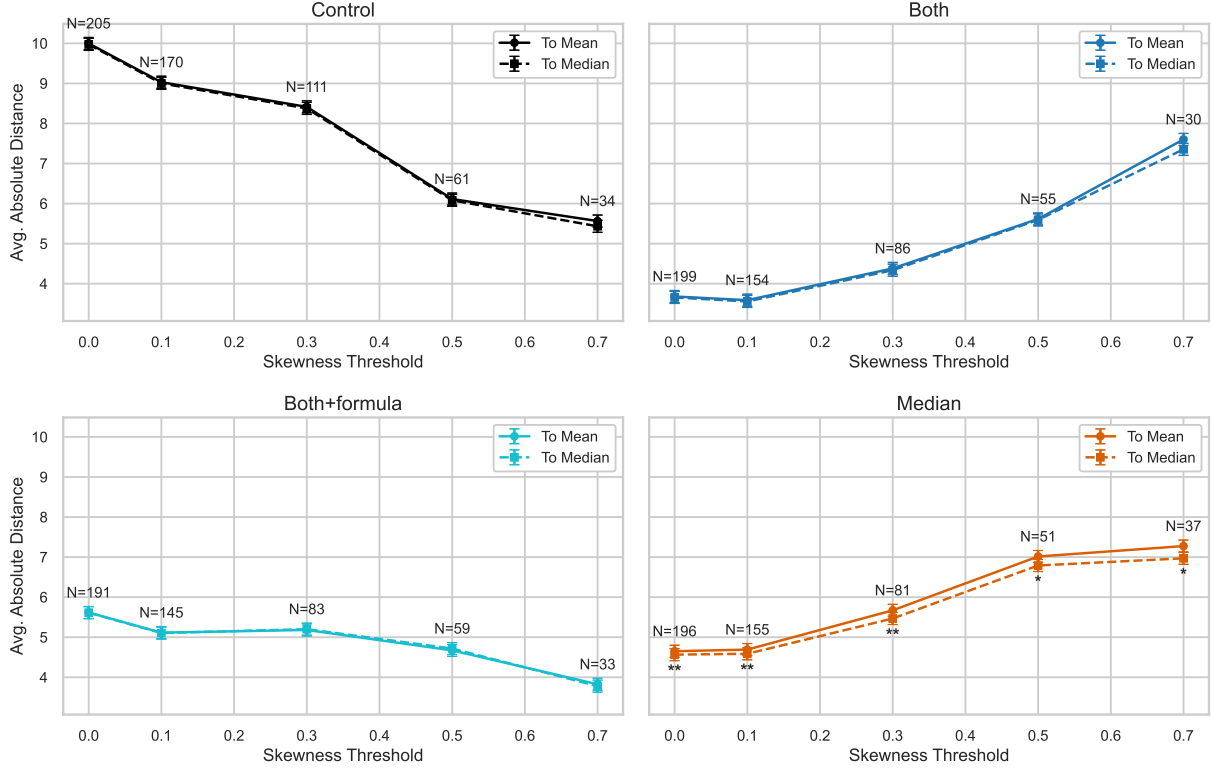


Figure 6: Point Forecast Distance to Mean vs. Median of Density Forecast

Notes: This figure shows the average absolute distance between respondents' point forecasts and the mean (solid line) or median (dashed line) of their density forecast, calculated for different levels of skewness. Each subplot corresponds to one treatment group. The x-axis reflects the skewness threshold used to classify distribution as "skewed," ranging from more inclusive (lower threshold) to more restrictive (higher threshold). Vertical error bars represent approximate 95% confidence intervals around group means. Stars indicate statistically significant differences between distances to mean and median based on paired t-tests: * $p < .1$, ** $p < .05$, *** $p < .01$.

4.4 Expectations and Spending

We also examine whether inflation point forecasts are consistent with respondents' expectations about future spending — that is, whether higher inflation forecasts are systematically associated with expectations of increased household consumption. This relationship serves as an external validity check, allowing us to evaluate whether different expectation measures capture behaviorally relevant beliefs and to identify which measures provide a more accurate reflection of economic behavior. To explore this relationship, we estimate the following regression:

$$E[C_i] = \theta_0 + \theta_1 E[\pi_i] + \mathbf{W}_i' \boldsymbol{\phi} + \varepsilon_i, \quad (9)$$

where $E[C_i]$ denotes expected household spending in 12 months, $E[\pi_i]$ is the inflation point forecast, and $\mathbf{W}_i$ is a vector of control variables.

Table 7 shows that the coefficient θ_1 is larger in the *Both* and *Both+Formula* treatment groups, suggesting that incentivized respondents exhibit stronger internal consistency between inflation expectations and planned consumption behavior. As expected, correlation between inflation and spending expectations is stronger if we look only at respondents who indicated that they considered price factor in their spending forecast. Further, we also find that respondents with higher inflation expectations in incentivized treatments are also more likely to select that they would make real changes to their spending. These findings provide further evidence that simple incentive structures can improve the economic coherence of survey responses.

	All				Price Factors			
	Control	Both	Both+formula	Median	Control	Both	Both+formula	Median
Point forecast	0.114 (0.08)	0.220* (0.12)	0.164** (0.08)	0.072 (0.09)	0.177* (0.09)	0.276*** (0.10)	0.179** (0.09)	0.125 (0.08)
Intercept	18.286*** (6.85)	-2.063 (3.31)	9.088 (6.91)	14.314** (6.78)	19.542** (9.35)	-4.366 (2.89)	13.092 (9.48)	19.042** (7.58)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	257	257	257	253	178	164	173	169
R^2	0.184	0.238	0.230	0.215	0.264	0.309	0.329	0.399

Table 7: Correlation Between Inflation Expectations and Nominal Spending

Notes: The table presents the relationship between expected inflation and expected spending. First four columns shows the results for entire dataset. The rest columns show results for the case then we restrict sample to only those observations where respondents associate change in spending with changes in future prices. Regressions are estimated by OLS with robust standard errors. Data are winsorized at the 1% and 99% levels. The control variables are age, race, gender, education, income, political affiliation, and primary grocery shopper status. * $p < .1$, ** $p < .05$, *** $p < .01$

5 External Validation and Measurement of Expectations

As additional validation, we compare our data with the data from the Survey of Consumer Expectations (SCE) across several dimensions and uncover a noteworthy pattern. The consistency between point forecasts and density forecasts among first-time SCE participants is systematically lower, resembling the responses of the unincentivized Control group in our experiment (Figure 7). In contrast, SCE participants with higher survey tenure exhibit substantially greater consistency, closely aligning with the incentivized groups in our study.

When examining the distribution of point forecasts in the SCE microdata from January to July 2024 (the most recent available), we find that the behavior of the Both treatment group in our experiment closely mirrors that of SCE respondents with 2–6 months of survey experience (Figure A-6).

These findings resonate with Kim and Binder (2023), who document strong “learning-through-survey” effects in the SCE, showing that greater survey tenure leads participants to become more attentive to inflation. They propose two potential explanations: (1) repeated participation increases real-world attention to inflation, and (2) lower-tenure respondents introduce more reporting error due to the cognitive demands of articulating their beliefs.

Our results lend stronger support to the second explanation — the *reporting error hypothesis*. This view suggests that respondents hold stable underlying beliefs about inflation, but those with low tenure struggle to report them accurately because doing so requires cognitive effort. Our findings reinforce this interpretation: while incentives did not substantively increase information search, they did meaningfully alter the distribution of forecasts. This suggests that higher incentives encouraged respondents to invest more cognitive effort into accurately retrieving and reporting their existing beliefs.

This interpretation has important implications for how we understand the dynamics of inflation expectations in nationally representative surveys such as the SCE and MSC. In par-

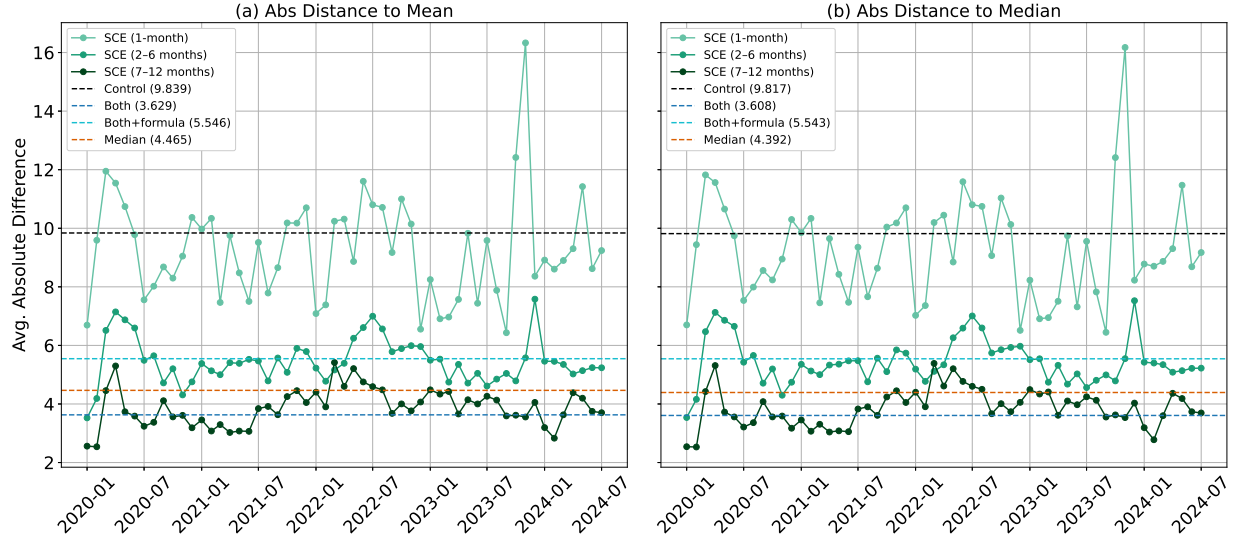


Figure 7: SCE Consistency For Different Tenure

ticular, it highlights the need to interpret changes in both the mean and dispersion (i.e., uncertainty and disagreement) of inflation expectations through the lens of rational inattention theory. Fluctuations in these moments may not necessarily reflect new information but rather changes in the costs or benefits of accurately reporting pre-existing beliefs. More importantly, our findings have implications for academic research, where analyses typically rely on a single survey wave and thus on noisier measures of expectations. In settings where measurement error is particularly consequential—such as quantitative assessments—it may be beneficial to introduce incentives, even for a subset of respondents, to obtain more accurate measures and draw more robust conclusions.

To more precisely capture the effect of information on belief updating, it may be useful to focus on respondents for whom the cognitive cost of reporting is lower or for whom the incentives to report accurately are higher. In practice, this includes individuals with greater survey experience, as they appear to face lower reporting costs. Thus, limiting analyses to more experienced respondents or designing experiments that provide strong incentives for accurately retrieving and reporting the existing beliefs may help isolate genuine updating behavior from noise introduced by reporting frictions. Alternatively, researchers can focus on subgroups of respondents who tend to exhibit lower reporting error even in the absence of

incentives. In [Table ??](#), we compare the distribution of point forecasts in the *Both* group to those of various demographic subgroups within the *Control* group. The results suggest that older, more educated, and higher-income respondents show the closest alignment, implying they may report their beliefs more accurately even without additional motivation. More generally, depending on the objectives of the study and whether capturing both attentive and inattentive expectations is important, the use of simple incentives may be desirable. This approach can provide a cost-effective alternative to repeated survey waves with the same respondents.

Group	Category	N	Wasserstein Distance	KS Statistic
Age	Over 60	58	6.97	0.20
	40 to 60	84	7.38	0.21
	Under 40	115	8.31	0.23
Education	College	139	5.16	0.16
	High School	34	8.12	0.15
	Some College	84	11.63	0.32
Income	Over 100k	79	7.27	0.19
	Under 50k	83	8.30	0.19
	50k to 100k	95	8.48	0.21

Table 8: Similarity Between Control Subgroups and Both Group Forecasts

6 Implications for Macroeconomic Models

Our findings have important implications for how inflation expectations are incorporated into macroeconomic models. A large literature uses survey-based measures of expectations to discipline models of expectation formation, disagreement, and information frictions (e.g., [Coibion and Gorodnichenko \(2015\)](#), [Mankiw and Reis \(2002\)](#), or [Sims \(2003\)](#)). If these measures are affected by survey design, the resulting estimates of key parameters may need to be adjusted to account for potential biases.

First, we find that incentivized expectations exhibit substantially lower means and reduced

upward bias. In New Keynesian models, average expected inflation plays a central role in determining inflation dynamics and policy trade-offs. Overstating inflation expectations can lead to an overestimation of inflation persistence or the strength of backward-looking components in hybrid Phillips curves (e.g., [Galí and Gertler \(1999\)](#), or [Coibion and Gorodnichenko \(2015\)](#)). As a result, models calibrated to unincentivized survey data may attribute excessive weight to adaptive expectations or misperceive the degree of anchoring.

Second, incentives significantly reduce cross-sectional disagreement. Disagreement is a key object in models with heterogeneous expectations, where it affects aggregate dynamics, price setting, and the transmission of monetary policy (e.g., [Mankiw et al. \(2003\)](#), [Angeletos and Lian \(2018\)](#), or [Bordalo et al. \(2020\)](#)). If survey data overstate disagreement due to reporting noise or inattention, models may overestimate the importance of expectation dispersion as a source of macroeconomic fluctuations or as a determinant of policy effectiveness.

Taken together, these findings suggest that survey design can influence the measurement of expectations in ways that affect macroeconomic inference. Incorporating attention-enhancing elicitation methods may therefore lead to more accurate estimates of expectation formation, disagreement, and learning, and ultimately improve the calibration and evaluation of macroeconomic models.

7 Conclusion

This paper provides new evidence on how incentive design shapes the measurement of inflation expectations in survey environments. Using a controlled experimental setting, we compare flat-fee elicitation with a range of marginal-incentive schemes that vary in their theoretical properties and cognitive complexity. Our results show that introducing performance-based incentives materially improves the quality of reported expectations: incentivized respondents exert greater effort, report fewer extreme values, exhibit lower upward bias, and produce beliefs that are more internally consistent and more closely aligned with both pro-

fessional benchmarks and economically meaningful behaviors.

Importantly, we also find that the behavioral effectiveness of an incentive scheme does not necessarily align with its theoretical sophistication. While strictly proper scoring rules guarantee truthful reporting under certain assumptions, their added complexity can limit their practical performance. In contrast, simple and transparent incentive schemes—despite lacking formal incentive compatibility—consistently perform as well as or better than more complex mechanisms across a wide range of validation criteria. This finding highlights a key trade-off between incentive compatibility and implementability: mechanisms that are easier to understand and use may better elicit meaningful beliefs in practice, particularly in large-scale surveys with heterogeneous and cognitively constrained respondents.

These results have important implications for both research and policy. For empirical macroeconomics, they suggest that widely used survey measures of expectations may contain nontrivial measurement error driven by weak incentives and reporting errors. Incorporating even simple performance-based incentives can substantially improve data quality and quantitative assessments, potentially leading to more accurate estimation and interpretation of models—especially those, such as New Keynesian frameworks, that rely heavily on expectations data. For policymakers, our findings underscore that the way expectations are measured—and not just how they are formed—can influence conclusions about expectation anchoring and heterogeneity.

More broadly, our study contributes to the growing literature on behavioral incentive compatibility by showing that cognitive constraints play a first-order role in belief elicitation. Future research could explore how to optimally balance simplicity, theoretical rigor and implementability convenience, how these findings extend to other domains of expectations, and how incentive schemes interact with information provision and learning over time. Taken together, our results suggest that improving the measurement of expectations requires not only better models of belief formation, but also careful attention to the design of the tools used to elicit them.

Bibliography

- Abeler, J., D. Huffman, and C. Raymond (2025). Incentive complexity, bounded rationality, and effort provision. *American Economic Review* 115(12), 4404–4437.
- Alfarano, S., E. Camacho-Cuena, A. Colasante, and A. Ruiz-Bufo (2023). The effect of time-varying fundamentals in learning-to-forecast experiments. *Journal of Economic Interaction and Coordination*, 1–29.
- Angeletos, G.-M. and C. Lian (2018). Forward guidance without common knowledge. *American Economic Review* 108(9), 2477–2512.
- Armantier, O., W. Bruine de Bruin, S. Potter, G. Topa, W. Van Der Klaauw, and B. Zafar (2013). Measuring inflation expectations. *Annu. Rev. Econ.* 5(1), 273–301.
- Armantier, O., G. Topa, W. Van der Klaauw, and B. Zafar (2017). An overview of the survey of consumer expectations. *Economic Policy Review* (23-2), 51–72.
- Bordalo, P., N. Gennaioli, Y. Ma, and A. Shleifer (2020). Overreaction in macroeconomic expectations. *American Economic Review* 110(9), 2748–2782.
- Carroll, C. D. (2001). The epidemiology of macroeconomic expectations.
- Cavallo, A., G. Cruces, and R. Perez-Truglia (2017). Inflation expectations, learning, and supermarket prices: Evidence from survey experiments. *American Economic Journal: Macroeconomics* 9(3), 1–35.
- Charness, G., U. Gneezy, and A. Imas (2013). Experimental methods: Eliciting risk preferences. *Journal of economic behavior & organization* 87, 43–51.
- Charness, G., U. Gneezy, and V. Rasocha (2021). Experimental methods: Eliciting beliefs. *Journal of Economic Behavior & Organization* 189, 234–256.
- Coibion, O. and Y. Gorodnichenko (2015). Information rigidity and the expectations formation process: A simple framework and new facts. *American Economic Review* 105(8), 2644–2678.

- Colasante, A., S. Alfarano, E. Camacho-Cuena, and M. Gallegati (2020). Long-run expectations in a learning-to-forecast experiment: a simulation approach. *Journal of Evolutionary Economics* 30, 75–116.
- Crosetto, P. and A. Filippin (2013). The “bomb” risk elicitation task. *Journal of risk and uncertainty* 47, 31–65.
- D’Acunto, F., U. Malmendier, and M. Weber (2023). What do the data tell us about inflation expectations? In *Handbook of economic expectations*, pp. 133–161. Elsevier.
- Danz, D., L. Vesterlund, and A. J. Wilson (2022). Belief elicitation and behavioral incentive compatibility. *American Economic Review* 112(9), 2851–2883.
- Danz, D., L. Vesterlund, and A. J. Wilson (2024). Evaluating behavioral incentive compatibility: Insights from experiments. *Journal of Economic Perspectives* 38(4), 131–154.
- De Bruin, W. B., W. Van der Klaauw, G. Topa, J. S. Downs, B. Fischhoff, and O. Armantier (2012). The effect of question wording on consumers’ reported inflation expectations. *Journal of Economic Psychology* 33(4), 749–757.
- Drobot, S., D. Puzzello, R. Rholes, and A. Wabitsch (2024). Incentivizing inflation expectations. *Available at SSRN 5226305*.
- D’Acunto, F., U. Malmendier, J. Ospina, and M. Weber (2021). Exposure to grocery prices and inflation expectations. *Journal of Political Economy* 129(5), 1615–1639.
- Engelberg, J., C. F. Manski, and J. Williams (2009). Comparing the point predictions and subjective probability distributions of professional forecasters. *Journal of Business & Economic Statistics* 27(1), 30–41.
- Evans, G. W., C. G. Gibbs, and B. McGough (2021). A unified model of learning to forecast. Technical report.
- Eyting, M. and P. Schmidt (2021). Belief elicitation with multiple point predictions. *European Economic Review* 135, 103700.

- Galí, J. and M. Gertler (1999). Inflation dynamics: A structural econometric analysis. *Journal of Monetary Economics* 44(2), 195–222.
- Gneiting, T. and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American statistical Association* 102(477), 359–378.
- Goldfayn-Frank, O., P. Kieren, and S. T. Trautmann (2024). A choice-based approach to the measurement of inflation expectations. Technical report, AWI Discussion Paper Series.
- Goldfayn-Frank, O., P. Kieren, and S. T. Trautmann (2025). A choice-based approach to the measurement of inflation expectations. *Available at SSRN 5233769*.
- Harrison, G. W., J. Martínez-Correa, J. T. Swarthout, and E. R. Ulm (2015). Eliciting subjective probability distributions with binary lotteries. *Economics Letters* 127, 68–71.
- He, B., L. I. Mostrom, and A. Sufi (2024). Investing in customer capital. Technical report, National Bureau of Economic Research.
- Hossain, T. and R. Okui (2013). The binarized scoring rule. *Review of Economic Studies* 80(3), 984–1001.
- Jonung, L. (1981). Perceived and expected rates of inflation in sweden. *The American Economic Review* 71(5), 961–968.
- Kim, G. and C. Binder (2023). Learning-through-survey in inflation expectations. *American Economic Journal: Macroeconomics* 15(2), 254–278.
- Krosnick, J. A. (1991). Response strategies for coping with the cognitive demands of attitude measures in surveys. *Applied cognitive psychology* 5(3), 213–236.
- Mankiw, N. G. and R. Reis (2002, November). Sticky information versus sticky prices: A proposal to replace the new keynesian phillips curve. *The Quarterly Journal of Economics* 117(4), 1295–1328.
- Mankiw, N. G., R. Reis, and J. Wolfers (2003). Disagreement about inflation expectations. *NBER Macroeconomics Annual* 18, 209–248.

- Marimon, R., S. E. Spear, and S. Sunder (1993). Expectationally driven market volatility: an experimental study. *Journal of Economic Theory* 61(1), 74–103.
- McMahon, M. and R. Rholes (2023). *Building central bank credibility: The role of forecast performance*. Centre for Economic Policy Research.
- Mokhtarzadeh, F. and L. Petersen (2021). Coordinating expectations through central bank projections. *Experimental economics* 24, 883–918.
- Palley, A. and S. Bansal (2021). Is it better to elicit quantile or probability judgments to estimate a continuous distribution? elicit both and create an inner crowd. *Elicit Both and Create an Inner Crowd*. (May 6, 2021). Kelley School of Business Research Paper (17-44).
- Petersen, L. and R. Rholes (2022). Macroeconomic expectations, central bank communication, and background uncertainty: A covid-19 laboratory experiment. *Journal of Economic Dynamics and Control* 143, 104460.
- Pfajfar, D. and E. Santoro (2008). Asymmetries in inflation expectation formation across demographic groups.
- Rholes, R. and L. Petersen (2021). Should central banks communicate uncertainty in their projections? *Journal of Economic Behavior & Organization* 183, 320–341.
- Rybacki, J. (2020). Macroeconomic forecasting in poland: The role of forecasting competitions. *Central European Economic Journal* 7(54), 1–11.
- Sims, C. A. (2003). Implications of rational inattention. *Journal of Monetary Economics* 50(3), 665–690.
- Smith, V. L. and J. M. Walker (1993). Monetary rewards and decision cost in experimental economics. *Economic inquiry* 31(2), 245–261.
- Xiao, W. (2024). Initial anchors and limited information in learning-to-forecast experiments. *Journal of Economic Behavior & Organization* 225, 192–227.

Appendix

A1 Other Tables and Figures

	Control (257 obs.)	Both (257 obs.)	Both+formula (257 obs.)	Median (253 obs.)
AGE:				
Under 40	44.7%	38.1%	38.9%	41.5%
40 to 60	32.7%	41.6%	43.2%	38.3%
Over 60	22.6%	20.2%	17.9%	20.2%
GENDER:				
Male	51.4%	48.2%	49.0%	49.8%
Female	47.9%	50.2%	50.6%	48.6%
Other	0.8%	1.6%	0.4%	1.6%
INCOME:				
Under 50k	32.3%	34.2%	33.5%	30.8%
50k to 100k	37.0%	35.0%	38.5%	34.4%
Over 100k	30.7%	30.7%	28.0%	34.8%
EDUCATION:				
High School	13.2%	12.1%	16.7%	13.0%
Some College	32.7%	34.2%	30.4%	30.4%
College	54.1%	53.7%	52.9%	56.5%
AFFILIATION:				
Republican	30.7%	27.2%	23.3%	23.3%
Independent	42.8%	47.1%	47.1%	48.2%
Democrat	26.5%	25.7%	29.6%	28.5%

Table A-1: Demographics: Sample Comparisons

Notes: This table presents the demographic composition of each treatment group. Values indicate the percentage share of respondents within each group.

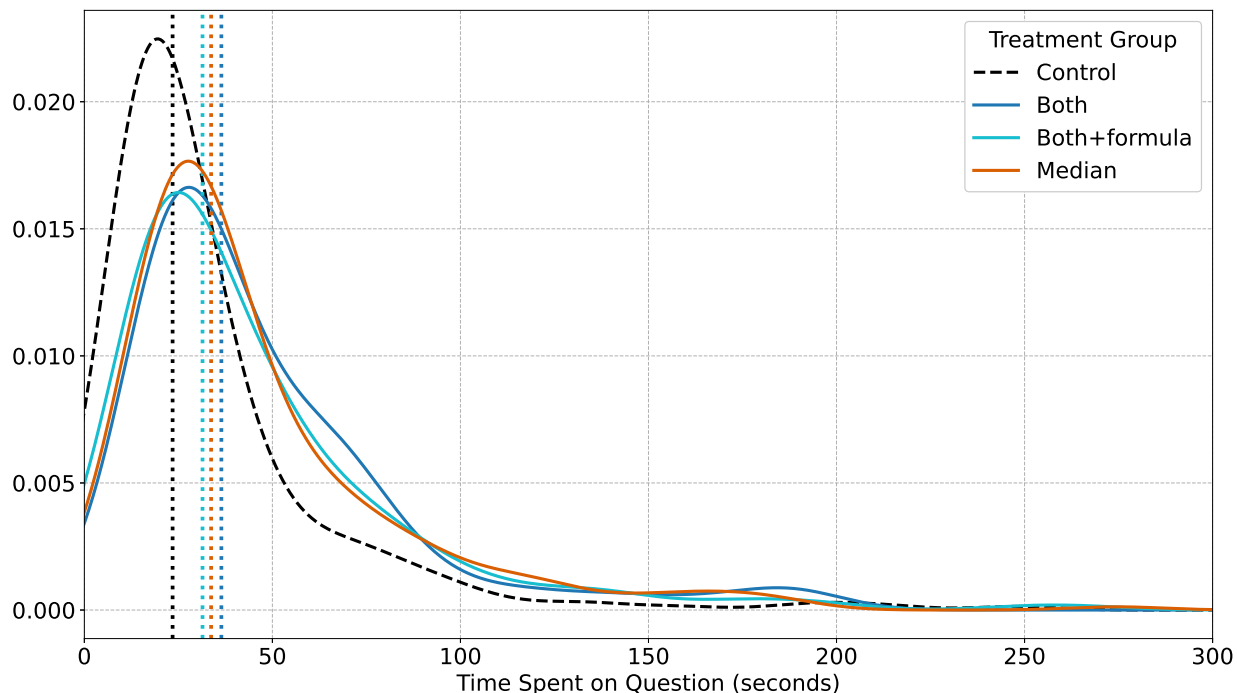


Figure A-1: Response Time Distribution to Point Forecast Question by Treatment Group

Notes: This figure shows the estimated probability density (PDF) of the time taken to answer the point inflation expectations question across different treatment groups. The vertical axis represents kernel density estimates, while the horizontal axis shows response time in seconds. To avoid distortion from extreme values, the top 1% of the longest response times were excluded. Each line corresponds to a treatment group. Vertical lines indicate medians.

A2 Textual Analysis of the Open-Ended Question

To better understand the intuition behind respondents' forecasts, we included the following open-ended question in our survey:

“Can you explain how you chose your estimate for the inflation rate over the next 12 months? What factors or considerations influenced your guess? Please share any thoughts, observations, or information that you took into account while providing your guess. Please type these in the box below.”

We begin by cleaning the responses using standard natural language processing techniques. Text was lowercased, and HTML entities, URLs, digits, punctuation, extra whitespace, and non-ASCII characters were removed. The cleaned text was then tokenized, and tokens consisting of non-alphabetic characters or English stopwords were excluded. Finally, the

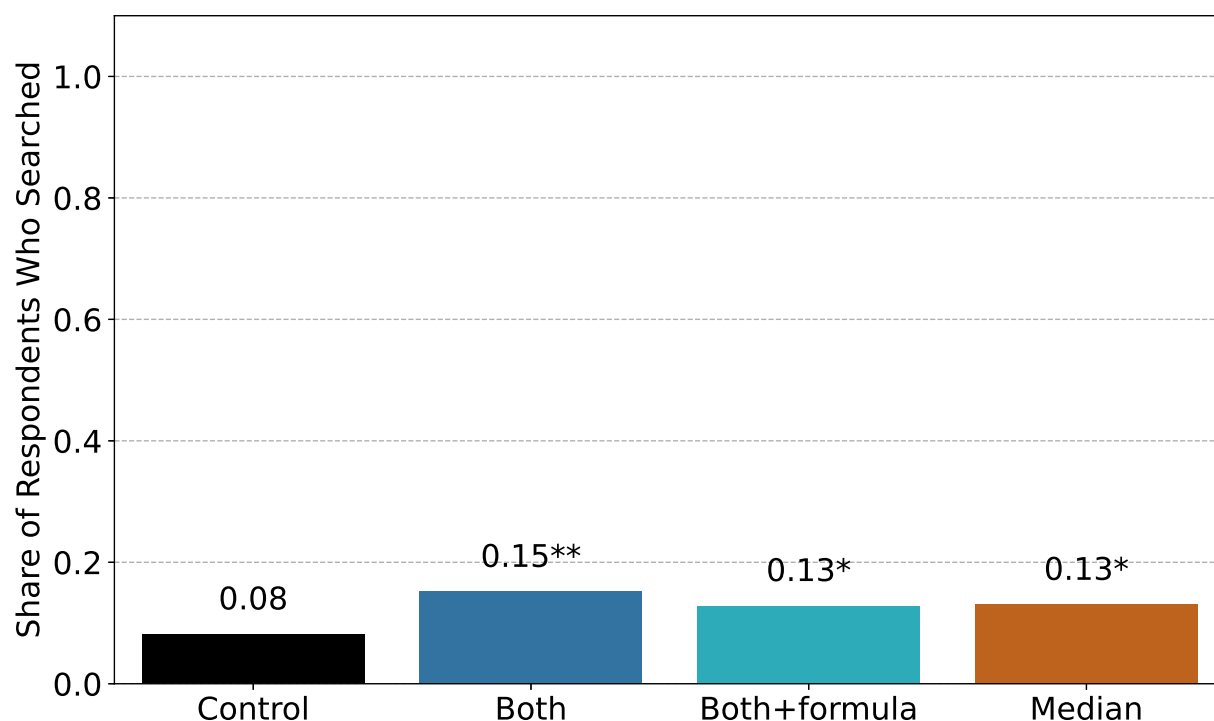


Figure A-2: Information Search by Treatment Group

Notes: This figure shows the share of respondents who reported searching for information about inflation online, broken down by treatment group. Respondents who answered "No" to the search question were coded as not having searched; all others were considered to have searched. Specifically, we asked "In providing your estimate for the inflation rate over the next 12 months, did you consult any source?" The vertical axis represents the proportion of searchers within each group. Asterisks indicate statistically significant differences from the *Control* group (z -test). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

remaining tokens were lemmatized using the WordNet lemmatizer to reduce words to their base forms.

To further analyze the content of the responses, we classified each one into one of four categories: *forward-looking*, *backward-looking*, *both*, or *unclear*. A response was classified as *forward-looking* if it contained terms such as "tariff," "supply_chain," "trump," "fed policy," "trade war," "current administration," or "president trump." It was labeled *backward-looking* if it included terms such as "trend," "past," "recent inflation," "current inflation," "inflation trend," or "last year." If a response included terms from both categories, it was labeled *both*; if it included none, it was classified as *unclear*.

Following the cleaning stage, we conducted an n-gram analysis to detect common multi-word phrases. Based on the results, we replaced some frequently occurring expressions with more

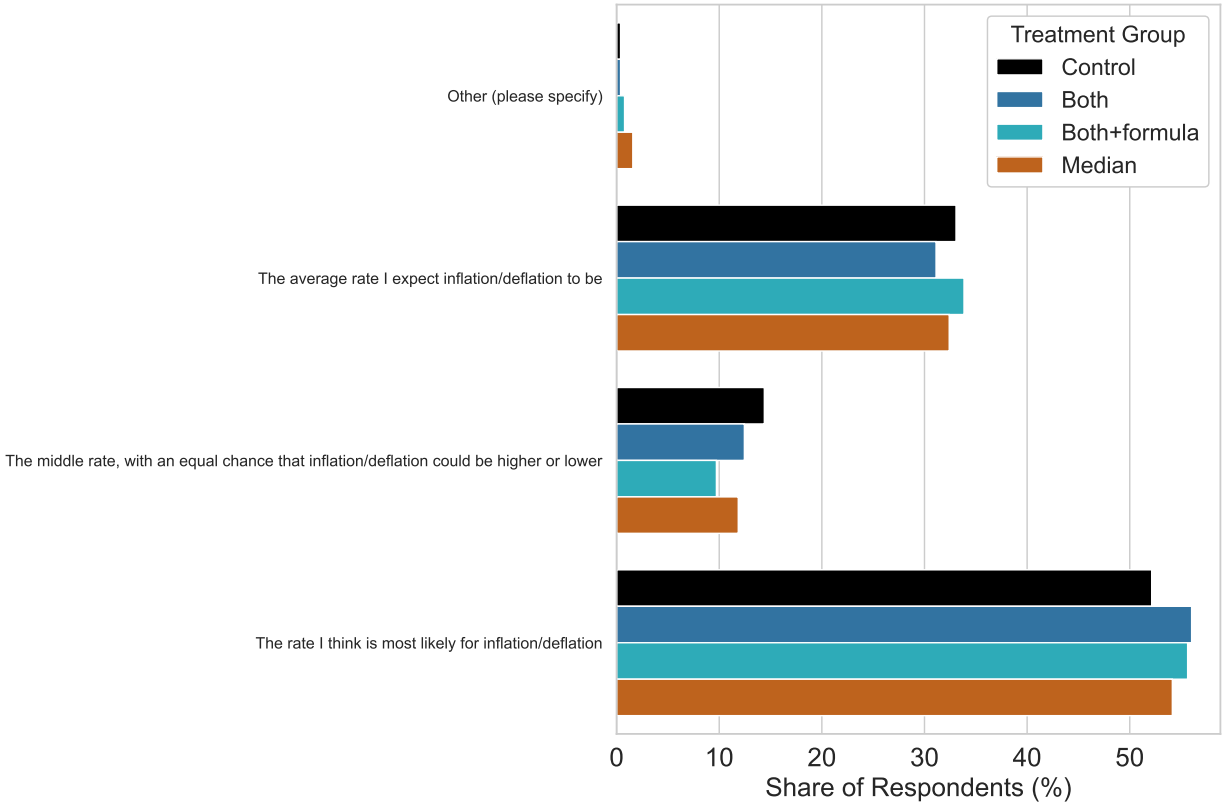


Figure A-3: Share of Respondents Choosing Mean, Median, or Mode

compact forms. For instance, “inflation rate” was replaced with “inflation,” “interest rate” with “interest_rate,” and both “federal reserve” and “central bank” with “fed.” We also removed the trigram “inflation next month,” as it simply repeats the survey question and does not offer informative content for analysis.

Next, we generated word clouds to visualize the most common words used in the cleaned responses. As shown in [Figure A-4](#), there are no substantial differences in the nature of respondents’ reasoning across treatment groups.

Table [A-5](#) presents the distribution of these categories across experimental groups. We do not observe substantial differences in the use of backward- or forward-looking language. The Control group has a slightly higher share of backward-looking responses, but overall patterns are similar across all groups. We also do not observe any substantial differences in the length of responses across treatment groups.

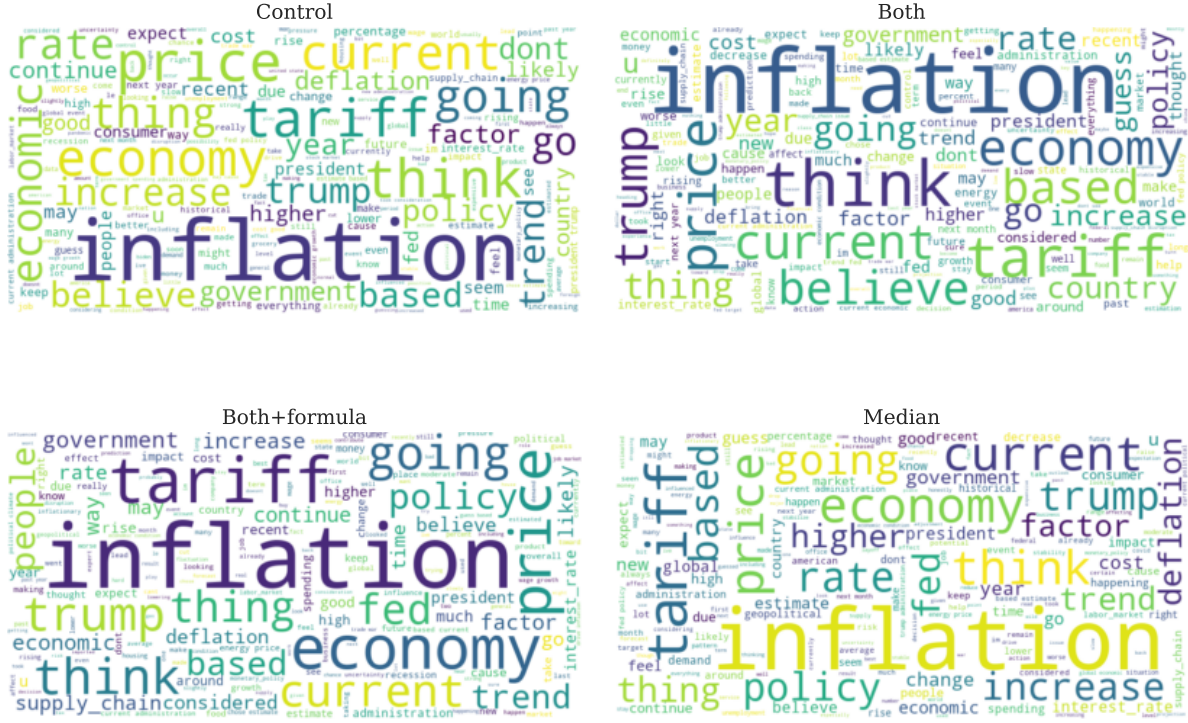


Figure A-4: Word Clouds by Treatment Group

As a robustness check, we reclassified the open-ended inflation forecast explanations using a large language model (LLM). Specifically, we used OpenAI’s GPT-4-turbo (ChatGPT, July 2024 version) to classify each response into one of three categories: Backward-looking, Forward-looking, or Unclear. The model was prompted with the following instruction:¹

”We are economists conducting research in the expectations formation of households. Your task is to read the following document (column TEXT) and determine the extent to which households use backwards-looking logic and forward-looking logic in explaining their inflation forecasts. Based on your reading of the doc-

¹In writing the prompt we follow [He et al. \(2024\)](#).

ument, please use your best judgment to classify their responses to one of two categories: backward-looking heuristics and forward-looking heuristics. Here is the document:”

We compare these results to our keyword-based heuristic classification and find broadly consistent patterns across treatment groups: responses tend to be more forward-looking overall, and there are no noticeably large differences between groups. The main distinction is that the LLM classification identifies the *Both* group as having the highest share of forward-looking responses, while the share of forward-looking responses in the other incentivized groups is lower compared to the *Control* group.

Exploring the simple distributional scoring rule

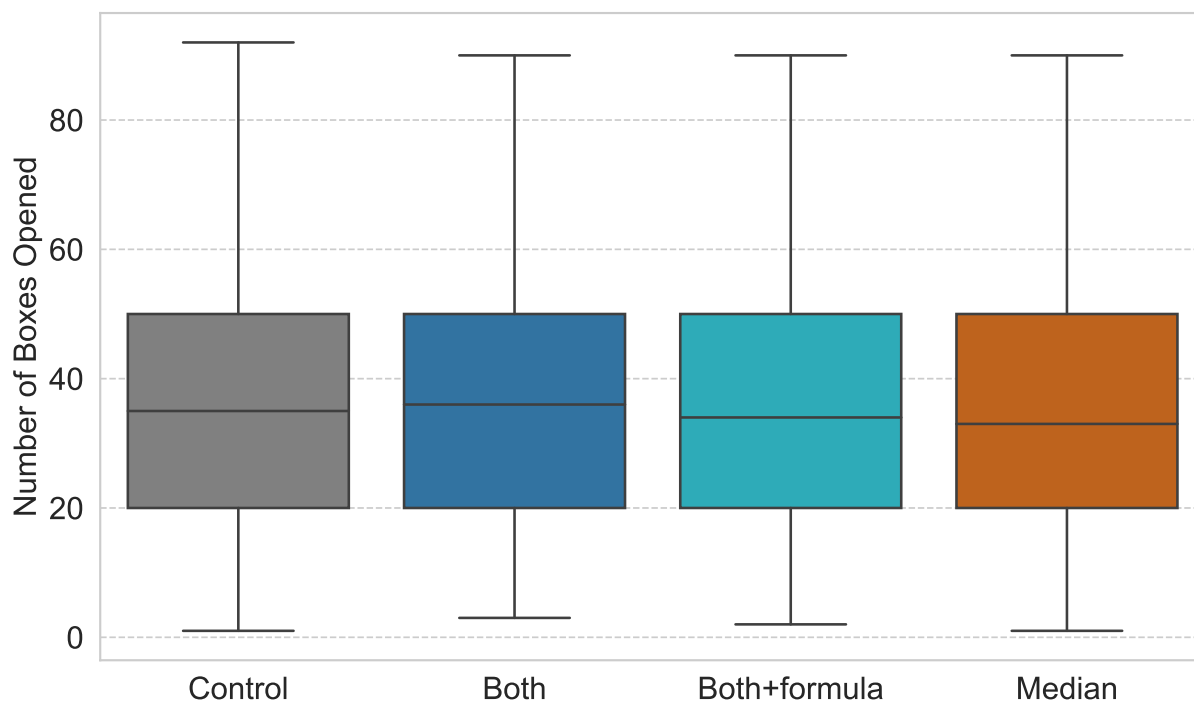


Figure A-5: Results of BRET Task

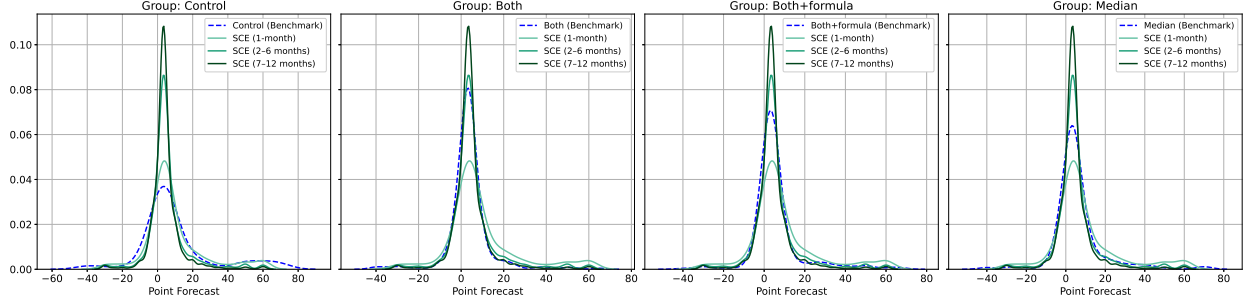


Figure A-6: SCE PDF For Different Tenure

Treatment	SPF Mean (Std. Dev.)	Treatment Mean (Std. Dev.)	Difference	Welch's t-stat	p-value
<i>Control</i>	2.30 (0.330)	10.00 (21.140)	7.70	5.831	0.000
<i>Both</i>	2.30 (0.330)	3.75 (10.305)	1.45	2.248	0.025
<i>Both+formula</i>	2.30 (0.330)	5.69 (12.328)	3.38	4.389	0.000
<i>Median</i>	2.30 (0.330)	5.82 (12.668)	3.51	4.402	0.000

Table A-2: Comparing Experimental Data to Professional Forecasts from SPF

Notes: This table compares data from the Survey of Professional Forecasters (SPF) to data from participants in the *Control*, *Both*, *Both+formula*, and *Median* groups using Welch's t-tests. The SPF data refer to the mean PCE inflation forecast for Q1 2025 to Q1 2026 (PCEB), based on the survey that concluded on February 11, 2025, which most closely aligns with our experimental time frame in March 2025. Note that the sample size for the SPF ($N = 37$) is considerably smaller than those in our survey; therefore, we use Welch's t-test to account for unequal variances and sample sizes. Treatment group data are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers.

Table A-3: Comparing Experimental Data to Professional Forecasts from SPF

	Mean Comparison (Welch's t-test p-values)				Variance Comparison (Levene's test p-values)			
	Control	Both	Both+formula	Median	Control	Both	Both+formula	Median
Control	Mean: 10.0				Var: 446.92			
Both	0.000***	Mean: 3.75			0.000***	Var: 106.2		
Both+formula	0.005***	0.054*	Mean: 5.69		0.000***	0.060*	Var: 151.97	
Median	0.007***	0.044**	0.906	Mean: 5.82	0.000***	0.016**	0.605	Var: 160.48

Notes: This table compares mean point forecasts and point forecast variances across treatment groups using Welch's t-tests and Levene's tests, respectively. The left panel reports mean point forecasts (diagonal) and p-values from Welch's t-tests for pairwise group comparisons (off-diagonal). The right panel shows the corresponding variances (diagonal) and p-values from Levene's test of equal variances (off-diagonal). Statistically significant result suggest that there is difference in mean/variance between two samples. Treatment group data are winsorized at the 1st and 99th percentiles to reduce the influence of outliers. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table A-4: Differences in Means and Variances Between Incentivized and Control Groups**(a) Three-Year Ahead Point Forecast**

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	4.21	6.85	-1.88	0.06	193.49	313.54	10.50	0.00
Both+formula	5.05	6.85	-1.29	0.20	184.21	313.54	8.81	0.00
Median	5.46	6.85	-1.01	0.31	170.57	313.54	7.52	0.01

(b) Perceived Inflation

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	5.64	7.89	-2.03	0.04	104.53	212.19	10.90	0.00
Both+formula	4.94	7.89	-2.70	0.01	94.35	212.19	18.39	0.00
Median	5.86	7.89	-1.76	0.08	129.20	212.19	7.91	0.01

(c) Gasoline Price Expectations

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	2.95	3.54	-0.58	0.56	129.85	133.60	0.10	0.75
Both+formula	3.78	3.54	0.23	0.82	138.96	133.60	0.00	0.98
Median	3.14	3.54	-0.39	0.69	134.38	133.60	0.00	0.97

(d) Food Price Expectations

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	5.42	6.87	-1.44	0.15	127.69	135.00	0.09	0.76
Both+formula	6.64	6.87	-0.22	0.82	139.58	135.00	0.05	0.82
Median	7.76	6.87	0.83	0.41	156.08	135.00	1.20	0.27

(e) Medical Care Price Expectations

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	8.80	10.00	-0.95	0.34	160.45	242.54	0.45	0.50
Both+formula	9.69	10.00	-0.23	0.82	196.60	242.64	0.15	0.69
Median	11.80	10.00	1.17	0.24	360.80	242.64	4.63	0.03

(f) Education Price Expectations

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	8.32	8.74	-0.33	0.74	215.07	200.95	0.01	0.92
Both+formula	9.88	8.74	0.92	0.36	203.79	200.95	0.61	0.43
Median	9.23	8.74	0.38	0.70	217.27	200.95	0.35	0.55

(g) Rent Price Expectations

	Mean				Variance			
	Mean	Control Mean	Test Statistic	p-value	Variance	Control Variance	Test Statistic	p-value
Both	8.77	10.67	-1.43	0.15	201.61	251.21	2.55	0.11
Both+formula	9.16	10.67	-1.21	0.23	144.34	251.21	6.53	0.01
Median	10.03	10.67	-0.48	0.63	189.79	251.21	1.88	0.17

Notes: The table presents the results of statistical tests comparing the incentivized groups to the non-incentivized *Control* group. Specifically, it reports Welch's *t*-test for differences in means and Levene's test for differences in variances. All data are winsorized at the 1st and 99th percentiles to mitigate the influence of outliers. * $p < .1$, ** $p < .05$, *** $p < .01$.

Table A-5: Summary of Open-Ended Responses by Treatment Group

Group	Length		LLM-based			Keyword-based			
	Mean Length	Median Length	Backward	Forward	Unclear	Backward	Both	Forward	Unclear
Control	18.61	15.0	0.202	0.595	0.202	0.125	0.078	0.366	0.432
Both	18.51	14.0	0.202	0.642	0.156	0.101	0.086	0.374	0.440
Both+formula	17.75	13.0	0.222	0.514	0.265	0.082	0.066	0.397	0.455
Median	18.67	14.0	0.174	0.510	0.316	0.067	0.071	0.403	0.458

Notes: The table reports the average and median number of tokens in responses to the open-ended question, along with the distribution of temporal focus classifications based on LLM and keywords (see Appendix A2 for details).

A3 Mathematical Proofs

A3.1 Incentive compatibility of the point prediction scoring function under risk neutrality

Proof. Given a guess g and a belief $f(i)$ regarding the density distribution of the inflation rate, a risk-neutral individual wants to maximize the expected payoff:

$$\int_{-\infty}^{\infty} a \times 2^{-|g-i|} f(i) di$$

The individual wants to find the guess g^* that maximizes the expected payoff

$$g^* = \arg \max_g \int_{-\infty}^{\infty} a \times 2^{-|g-i|} f(i) di$$

Since a is positive, this is equivalent to finding the guess that minimizes the expected loss

$$g^* = \arg \min_g \int_{-\infty}^{\infty} |g-i| f(i) di$$

Since $|g-i|$ is a convex function on g , the following first order condition is sufficient for global minimum

$$\int_{-\infty}^{\infty} \frac{g^* - i}{|g^* - i|} f(i) di = 0, \quad \text{for } |g^* - i| \neq 0$$

Since $\frac{g^* - i}{|g^* - i|} = 1$ when $g^* > i$ and $\frac{g^* - i}{|g^* - i|} = -1$ when $g^* < i$, we can re-write the above condition as

$$\int_{g^*}^{\infty} f(i) di - \int_{-\infty}^{g^*} f(i) di = 0$$

Since $\int_{-\infty}^{\infty} f(i) di = 1$, then

$$1 - \int_{g^*}^{\infty} f(i)di - \int_{g^*}^{\infty} f(i)di = 0$$

$$\implies 1 = 2 \times \int_{g^*}^{\infty} f(i)di$$

$$\implies \int_{g^*}^{\infty} f(i)di = 0.5$$

Hence, the median of the belief distribution is the optimal report. □

A3.2 ?? is not incentive compatible for eliciting interval probabilities

Proof. Given a set of n bins $\{b_j\}_{j=1}^n$, where $\cup_{j=1}^n b_j = \mathcal{R}$ and $b_j \cap b_k = \emptyset$ for all $k \neq j$, a set of guesses $\{g_j\}_{j=1}^n$ where each element represents a guess on the probability that the inflation rate falls in the corresponding bin j , and a belief $f(i)$ regarding the density distribution of the inflation rate, a risk-neutral individual wants to maximize the expected payoff given by:

$$\begin{aligned} & \int_{-\infty}^{\infty} 10 \sum_{j=1}^n g_j \mathbb{1}(i \in b_j) f(i) di \\ & \text{s.t.} \quad \sum_{j=1}^n g_j = 1 \end{aligned}$$

Note that the objective function and the constraint are linear, so a solution satisfying the first order conditions maximizes the expected payoff. The Lagrangian for this problem is

$$\mathcal{L} = \int_{-\infty}^{\infty} 10 \sum_{j=1}^n g_j \mathbb{1}(i \in b_j) f(i) di - \lambda \sum_{j=1}^n g_j - 1$$

The first order conditions are:

$$10 \int_{\underline{b_j}}^{\overline{b_j}} f(i) di = \lambda \quad \forall j = 1, \dots, n$$

$$\sum_{i=1}^n g_j = 1$$

where $\underline{b_j}$ and $\overline{b_j}$ represent the lower and upper bin of bin b_j , respectively. Consider two arbitrary bins, b_1 and b_2 , for which the subject assigns different subjective probabilities. The first order conditions require $10 \int_{\underline{b_1}}^{\overline{b_1}} f(i) di = \lambda = 10 \int_{\underline{b_2}}^{\overline{b_2}} f(i) di$, a contradiction. Hence, the problem has no solution unless all bins have the same subjective probability. If all bins have the same subjective probability, any solution where the guesses sum up to one satisfies the first order conditions. $\square$

A3.3 Incentive compatibility of the generalized QSR for interval probabilities under risk neutrality

Proof. Given a set of n bins $\{b_j\}_{j=1}^n$, where $\cup_{j=1}^n b_j = \mathcal{R}$ and $b_j \cap b_k = \emptyset$ for all $k \neq j$, a set of guesses $\{g_j\}_{j=1}^n$ where each element represents a guess on the probability that the inflation rate falls in the corresponding bin j , and a belief $f(i)$ regarding the density distribution of the inflation rate, a risk-neutral individual wants to maximize the expected payoff given by:

$$\int_{-\infty}^{\infty} a + b \sum_{j=1}^n g_j \mathbb{1}(i \in b_j) - a \sum_{j=1}^n g_j^2 f(i) di$$

$$s.t. \quad \sum_{i=1}^n g_j = 1$$

Note that the objective function is strictly concave while the constraint is linear, so the solution satisfying the first order conditions uniquely maximizes the expected payoff. The Lagrangian for this problem is

$$\mathcal{L} = \int_{-\infty}^{\infty} a + b \sum_{j=1}^n g_j \mathbb{1}(i \in b_j) - a \sum_{j=1}^n g_j^2 f(i) di - \lambda \sum_{i=1}^n g_j - 1$$

The first order conditions are:

$$b \int_{\underline{b_j}}^{\overline{b_j}} f(i) di - 2a g_j = \lambda \quad \forall j = 1, \dots, n$$

$$\sum_{j=1}^n g_j = 1$$

where $\underline{b_j}$ and $\overline{b_j}$ represent the lower and upper bin of bin b_j , respectively. Since $b = 2a$ we can re-write the first order conditions as

$$\int_{\underline{b_j}}^{\overline{b_j}} f(i) di - g_j = \frac{\lambda}{b} \quad \forall j = 1, \dots, n$$

$$\sum_{j=1}^n g_j = 1$$

If $\lambda > 0$, then $\int_{\underline{b_j}}^{\overline{b_j}} f(i) di > g_j \forall j$ and since $\int_{-\infty}^{\infty} f(i) di = 1$ this implies $\sum_{j=1}^n g_j < 1$. By a similar logic, $\lambda < 0$ implies $\sum_{j=1}^n g_j > 1$. Hence, the optimum requires $\lambda = 0$ and

$$\int_{\underline{b_j}}^{\overline{b_j}} f(i) di = g_j \quad \forall j = 1, \dots, n$$

That is, the guess assigned to each interval is equal to the subjective probability that the inflation rate falls in the corresponding interval. $\square$

A3.4 Risk Aversion and Simple Distributional Elicitation Incentive Scheme

A3.4.1 Environment

Let $\{b_k\}_{k=1}^K$ denote a collection of disjoint bins that partition the outcome space $\mathcal{R}$ of the random variable π . A subject's true subjective belief is the probability vector

$$\mathbf{p} = (p_1, \dots, p_K) \in \Delta^{K-1}, \quad p_k = \Pr(\pi \in [\underline{b_k}, \overline{b_k}]),$$

where Δ^{K-1} is the $(K-1)$ -dimensional probability simplex. The subject reports a probability vector

$$\mathbf{g} = (g_1, \dots, g_K) \in \Delta^{K-1}$$

Under our simple *pay-the-probability* rule, a subject's payoff is

$$x = c g_k \quad \text{if } (\pi \in [\underline{b}_k, \overline{b}_k]), \quad c > 0 \text{ known.}$$

Preferences are represented by constant-relative-risk-aversion (CRRA) utility:

$$U(x) = \begin{cases} \frac{x^{1-\rho}}{1-\rho}, & \rho \neq 1, \\ \ln x, & \rho = 1, \end{cases} \quad \rho \geq 0,$$

where $\rho = 0$ corresponds to risk neutrality and $\rho > 0$ to risk aversion. We aim to characterise how risk preferences map true beliefs $\mathbf{p}$ into optimal reports $\mathbf{g}^*$, quantify the induced distortion, and show how (and when) the mapping can be inverted to recover $\mathbf{p}$ from observed reports, $\mathbf{g}$.

A3.4.2 Expected-utility maximisation

The agent chooses $\mathbf{g}$ to solve

$$\max_{\mathbf{g}} \sum_{k=1}^K p_k U(c g_k) \quad \text{s.t.} \quad \sum_{k=1}^K g_k = 1, \quad g_k \geq 0.$$

Define the Lagrangian

$$\mathcal{L} = \sum_{k=1}^K p_k U(c g_k) - \lambda \left(\sum_{k=1}^K g_k - 1 \right).$$

A3.4.3 First-order conditions

For each k with $g_k > 0$,

$$p_k U'(c g_k) c = \lambda.$$

Because $U'(x) = x^{-\rho}$ when $\rho \neq 1$ and $U'(x) = 1/x$ when $\rho = 1$, we may write the FOC for all $\rho > 0$ as

$$p_k c^{-\rho} g_k^{-\rho} c = p_k c^{1-\rho} g_k^{-\rho} = \lambda,$$

which we rewrite as

$$p_k g_k^{-\rho} = \frac{\lambda}{c^{1-\rho}}.$$

Because the right-hand side $\frac{\lambda}{c^{1-\rho}}$ is *the same constant for every k* , the stationarity condition

$$p_k g_k^{-\rho} = \frac{\lambda}{c^{1-\rho}} \tag{A0}$$

implies that all ratios $p_k g_k^{-\rho}$ are equal. Rearranging (A0) gives

$$g_k^\rho = \left(\frac{c^{1-\rho}}{\lambda}\right) p_k \implies g_k = \kappa p_k^{1/\rho}, \quad \kappa := \left(\frac{c^{1-\rho}}{\lambda}\right)^{1/\rho}.$$

The scalar κ is identical across bins; impose the simplex constraint to pin it down:

$$1 = \sum_{j=1}^K g_j = \kappa \sum_{j=1}^K p_j^{1/\rho} \implies \kappa = \frac{1}{\sum_{j=1}^K p_j^{1/\rho}}.$$

Substituting κ back yields the unique interior optimum

$$\boxed{g_k^* = \frac{p_k^{1/\rho}}{\sum_{j=1}^K p_j^{1/\rho}}, \quad \rho > 0.} \tag{A1}$$

Hence, for every $k \in K$, $g_k \propto p_k^{1/\rho}$; the multiplier λ (via κ) merely ensures the probabilities

re-scale to sum to one

Risk-neutral case ($\rho \rightarrow 0$).

This focuses on non-degenerate cases. For degenerate cases, a risk-neutral agent reports truthfully.

When $\rho = 0$, we have that $U(x) = x$. Given this, we have

$$\max_{\mathbf{g} \in \Delta^{K-1}} c \sum_{k=1}^K p_k g_k \implies \max_{\mathbf{g} \in \Delta^{K-1}} \sum_{k=1}^K p_k g_k,$$

so the constant c is irrelevant. Let

$$m = \max_j p_j, \quad M = \{k : p_k = m\} \quad (|M| \geq 1).$$

Intuitively, we are defining m as the highest truthful probability an agent assigns to any of the K disjoint bins. This is the modal bin. Using this, we build a modal set, M , comprising bins to which an agent truthfully assigns the outcome probability m . The cardinality condition says there must be a modal bin that is not necessarily unique.

Using this, we can rewrite each numerator of the interior solution (A1) as

$$p_k^{1/\rho} = m^{1/\rho} \left(\frac{p_k}{m} \right)^{1/\rho}.$$

If $k \in M$ the bracketed term equals 1; if $k \notin M$ it equals $e^{\left[\frac{\ln(\frac{p_k}{m})}{\rho} \right]} \xrightarrow{\rho \rightarrow 0^+} 0$. To see this, note that the numerator is negative, since $\frac{p_k}{m} < 1$ for any bin $k \notin M$. Since that is true, then the exponent goes to $-\infty$ and $\rho \rightarrow 0^+$.

The denominator is:

$$\begin{aligned}
Z(\rho) &:= \sum_{j=1}^K p_j^{1/\rho} \\
&= \underbrace{\sum_{j \in M} p_j^{1/\rho}}_{\text{modal bins}} + \underbrace{\sum_{j \notin M} p_j^{1/\rho}}_{\text{non-modal bins}} \\
&= m^{1/\rho} \left[\sum_{j \in M} \left(\frac{p_j}{m} \right)^{1/\rho} + \sum_{j \notin M} \left(\frac{p_j}{m} \right)^{1/\rho} \right] \\
&= m^{1/\rho} \left[\sum_{j \in M} 1 + \sum_{j \notin M} \left(\frac{p_j}{m} \right)^{1/\rho} \right] \\
&= m^{1/\rho} \left[|M| + \sum_{j \notin M} \left(\frac{p_j}{m} \right)^{1/\rho} \right].
\end{aligned}$$

Taking the limit of this:

$$m^{1/\rho} \left[|M| + \sum_{j \notin M} \left(\frac{p_j}{m} \right)^{1/\rho} \right] \xrightarrow{\rho \rightarrow 0^+} |M| m^{1/\rho}.$$

Dividing, we obtain the limit of the interior solution for a risk-neutral agent:

$$g_k^*(\rho) \xrightarrow{\rho \rightarrow 0^+} \begin{cases} \frac{1}{|M|}, & k \in M, \\ 0, & k \notin M. \end{cases}$$

Risk-neutral indifference. For any report $\mathbf{g}$ that puts *all* probability on the modal set M ,

$$\mathbb{E}[X] = c \sum_{k=1}^K p_k g_k = c m \sum_{k \in M} g_k = c m,$$

so the agent is indifferent among the class

$$\mathcal{G}_{\text{RN}} = \{ \mathbf{g} \in \Delta^{K-1} : \sum_{k \in M} g_k = 1, g_k = 0 \text{ for } k \notin M \}.$$

The equal-split limit in interior solution for a risk-neutral agent is simply one convenient

representative of this indifference class.

Log-utility case ($\rho = 1$). Setting $\rho = 1$ in the interior formula (A1) gives

$$g_k^*(1) = \frac{p_k^{1/1}}{\sum_{j=1}^K p_j^{1/1}} = \frac{p_k}{\sum_{j=1}^K p_j} = p_k \quad (k = 1, \dots, K),$$

because $\sum_j p_j = 1$ by definition of a probability vector. Hence, log utility delivers **exactly truthful reporting**. It is the threshold for ρ at which our pay-the-probability rule becomes strictly proper.

*Comparative statics in ρ

Starting from the interior solution

$$g_k^*(\rho) = \frac{p_k^{1/\rho}}{Z(\rho)}, \quad Z(\rho) = \sum_{j=1}^K p_j^{1/\rho}, \quad (\text{C1})$$

define the ρ -dependent weights $w_j(\rho) = \frac{p_j^{1/\rho}}{Z(\rho)}$ so that $\sum_j w_j(\rho) = 1$.

Taking logs and differentiating gives

$$\ln g_k^* = \frac{1}{\rho} \ln p_k - \ln Z(\rho), \quad \implies \quad \frac{\partial}{\partial \rho} \ln g_k^* = -\frac{1}{\rho^2} \ln p_k - \frac{Z'(\rho)}{Z(\rho)}. \quad (\text{C2})$$

In compact form, differentiating $Z(\rho) = \sum_j p_j^{1/\rho}$ gives

$$Z'(\rho) = -\frac{1}{\rho^2} \sum_j p_j^{1/\rho} \ln p_j = -\frac{1}{\rho^2} Z(\rho) \overline{\ln p}(\rho),$$

where $\overline{\ln p}(\rho) := \sum_j w_j(\rho) \ln p_j$ is the $w_j(\rho)$ -weighted mean of log-probabilities.

Sign of $\partial g_k^* / \partial \rho$. Substituting into (C2),

$$\frac{\partial}{\partial \rho} \ln g_k^* = -\frac{1}{\rho^2} [\ln p_k - \overline{\ln p}(\rho)].$$

Because the prefactor $-1/\rho^2 < 0$,

$$\boxed{\text{sign } \frac{\partial g_k^*}{\partial \rho} = \text{sign}\{\overline{\ln p}(\rho) - \ln p_k\}.}$$

- If $\ln p_k > \overline{\ln p}(\rho)$ (typically $p_k > 1/K$), then $\partial g_k^*/\partial \rho < 0$: high-probability bins lose weight as ρ rises.
- If $\ln p_k < \overline{\ln p}(\rho)$ (typically $p_k < 1/K$), then $\partial g_k^*/\partial \rho > 0$: low-probability bins gain weight as ρ rises.

In summary, the optimal report moves smoothly from corner solutions at $\rho \rightarrow 0^+$, to truthful reporting at $\rho = 1$, and finally to a uniform distribution as $\rho \rightarrow \infty$. This highlights log utility as the knife-edge where the pay-the-probability rule becomes strictly proper.

A3.4.4 Invertibility (recovering beliefs from reports)

(A1) is bijective for any fixed $\rho > 0$. Given an observed report $\mathbf{g}$ and known ρ , the latent beliefs are recovered by

$$p_k = \frac{g_k^\rho}{\sum_{j=1}^K g_j^\rho}, \quad k = 1, \dots, K. \quad (\text{A2})$$

A3.4.5 Distortion metric

Define the total variation (L_1) distortion between true beliefs and reported probabilities:

$$D(\rho) = \frac{1}{2} \sum_{k=1}^K |g_k^*(\rho) - p_k|.$$

Maximal distortion at $\rho = 0$. Let $m = \max_k p_k$ and let $M = \{k : p_k = m\}$ denote the modal set. If a risk-neutral agent puts all weight on M , then

$$\begin{aligned} D(0) &= \frac{1}{2} \sum_{k=1}^K |g_k^*(0) - p_k| \\ &= \frac{1}{2} \left[(1 - m) + \sum_{k \notin M} p_k \right] \\ &= 1 - m. \end{aligned}$$

Thus the worst-case distortion is

$$\boxed{D_{\max} = 1 - \max_k p_k},$$

whether the mass is placed on a single modal bin or split across ties.

Minimal distortion at $\rho = 1$. At log utility, $g_k^*(1) = p_k$ for all k , so

$$D(1) = 0.$$

Distortion otherwise. For $0 < \rho < \infty$, the derivative satisfies

$$D'(\rho) < 0 \text{ for } 0 < \rho < 1, \quad D'(\rho) > 0 \text{ for } \rho > 1.$$

Hence $D(\rho)$ decreases on $(0, 1]$, attains its minimum at $\rho = 1$, and increases on $[1, \infty)$, converging to

$$\lim_{\rho \rightarrow \infty} D(\rho) = \frac{1}{2} \sum_{k=1}^K \left| p_k - \frac{1}{K} \right|,$$

the distance between the true distribution and the uniform distribution.

Intuitively, distortion is largest when agents are risk-neutral (corner reports), vanishes under log utility (truthful reports), and grows again with extreme risk aversion as reports flatten toward uniformity.